

The impact of the Shanghai – Hong Kong stock market connection on corporate innovation: Evidence from mainland China

Abstract: The Shanghai stock market and the Hong Kong stock market were connected by the Chinese government in 2014, allowing Hong Kong investors to trade on a group of stocks on the Shanghai stock market. Using a difference-in-differences approach, we examine how this stock market connection affects corporate innovation in mainland China. We argue that the stock market connection plausibly facilitates external financing, enhances the informational feedback effect of stock prices, involves more sophisticated investors' monitoring and advising on firm management, and thereby spurs corporate innovation. Our results support this argument. We find that corporate innovation increases after the establishment of the stock market connection, and that this increase is larger for financially constrained firms, firms with highly informative stock prices, and firms that feature more Hong Kong institutional investments. We further show that the positive effect of the stock market connection on innovation is more pronounced for non-state-owned firms, firms with few political connections, firms with weak intellectual property rights protection, or firms that are headquartered in non-high-tech economic zones. Our study sheds light on how the opening of a developing stock market to a more developed stock market shapes corporate innovation.

Keywords: stock market connection; corporate innovation; external financing; informational content of stock prices, monitoring and advising on innovation

1. Introduction

Corporate innovation is widely regarded to be a major engine of a country's long-term economic growth and is vital for a country to maintain its competitive advantage (e.g., Solow, 1957; Romer, 1986). Motivating and nurturing corporate innovation is a pressing task for developing countries where technological innovation is relatively less developed. Empirical evidence from developed countries shows that varied economic environments and firm characteristics promote firm innovation. For instance, corporate venture capital (Chemmanur, Loutskina, and Tian, 2014), hedge fund activism (Brav, Jiang, Ma, Tian, 2018), credit supply (Amore, Schneider, and Zaldokas, 2013), private equity ownership (Ferreira, Manso, and Silva, 2014), board independence (Balsmeier, Fleming, Manso, 2016), intellectual property rights protection (Fang, Lerner, and Wu, 2017), managerial competencies (e.g., Custodio, Ferreira, and Matos, 2019; Chemmanur, Kong, Drishnan, and Yu, 2019), and long-term incentive plans on executive compensation (Manso, 2011) help enhance corporate innovation. Nonetheless, evidence on how economic policy and regulatory reform spur corporate innovation is sparse, especially in developing countries such as China, where there is considerable need to innovate to promote sustainable economic growth. Understanding the role of regulatory intervention in shaping corporate innovation is important, because a regulatory policy, if effective in stimulating the aggregate innovation activities in an economy, would largely promote a developing country's economic development.

In this study, we focus on a regulatory reform on an emerging stock market, because nurturing corporate innovation effectively requires a better-functioning stock market that allocates resources efficiently to facilitate firm investments and innovation. Specifically, we investigate whether the Shanghai-Hong Kong stock market connection program implemented by the Chinese government would enhance innovation of Chinese listed companies. In 2014, the stock market connection program commenced, allowing Hong Kong investors to trade on

the connected stocks. These consist of constituent stocks of the Shanghai Stock exchange 180 index and 380 index, and of stocks listed on both the Shanghai Stock Exchange and the Hong Kong Stock Exchange. As a consequence, more sophisticated market participants such as institutional investors from the Hong Kong stock market enter the Shanghai stock market to trade on the connected stocks. Since the Hong Kong stock market is regarded as better-developed than the Shanghai stock market (e.g., Ke, Rui, and Wei, 2012; Hung, Wong, and Zhang, 2012; Ke, Lennox, and Xin, 2015), the implementation of the Shanghai-Hong Kong stock market connection program in 2014 provides a unique setting to examine how the opening of a developing stock market to a more developed stock market can affect corporate innovation.

We argue that the stock market connection program could influence corporate innovation through three possible channels. First, in seeking external financing from the more developed Hong Kong stock market, the stock market connection increases and diversifies the sources of external funds for the connected Chinese firms, reducing the financing costs and financial risk of pursuing innovative projects. Stimulating innovation requires a large amount of funds, inclusive of immediate funds, that mostly need to be secured via external financing (e.g., Bena and Li, 2014; Ferreira, Manso, and Silva, 2014; Bernstein, 2015). The stock market connection program that facilitates external financing can mitigate financial risk or constraints that arise during the process of research and development for innovation. Thereby, the program can help accelerate the research and development progress in a way that leads to earlier success in innovation, and bring new, innovative products to the market earlier. As such, firms can better develop their competitive advantage in the product market competition and earn higher payoffs from innovation. These potential outcomes motivate and nurture firm innovation.

Second, as innovation is riskier, more idiosyncratic with more contingencies, than routine tasks (Holmstrom, 1989), the potential and outcome of innovative projects are hard to assess,

often resulting in investors holding divergent opinions about the prospects of the projects (Allen and Gale, 1999). The stock market connection program enhances the informational feedback effects of stock prices. Stock prices in a developed stock market may contain valuable information about the prospects of firms' investments and innovation (e.g., Hsu, Tian, and Xu, 2014). Consistent with this notion, a vast literature studying the U.S. stock market (e.g., Chen, Goldstein, and Jiang, 2007; Bakke and Whited, 2010; Dutta and Reichelstein, 2003; 2005; Foucault and Fresard, 2012, 2014; Loureiro and Taboada, 2015) provides evidence that managers account for information on stock prices and actively incorporate it into their investment decisions. Since the establishment of the stock market connection, more-sophisticated investors from the Hong Kong stock market get involved in stock trades, plausibly making stock prices more efficient and more informative about the prospects of innovative research and investments. This may help research and development to progress more efficiently for innovation outputs and thereby spur corporate innovation.

Third, innovation involves a long process that is fraught with uncertainty and has a high risk of failure in the short run (Holmstrom, 1989; He and Tian, 2013). However, due to compensation and career concerns, managers (especially those in technology-intensive industries) tend to sacrifice long-term sustainability for short-term performance goals (e.g., Graham, Harvey, and Rajgopal, 2005). Also, managers' rent extraction may reduce corporate cash holdings or other resources that are required in pursuing innovative projects. Therefore, managers' myopia and rent extraction stifle corporate innovation. These are more salient in China where the market for financial analysts is under-developed and institutional stock ownership is low (e.g., Chen, Ke, and Yang, 2013), but are less of a concern in Hong Kong where the stock market, in comparison, is more developed and mature (e.g., Hung, Wong, and Zhang, 2012; Ke, Rui, and Yu, 2012; Ke, Lennox, and Xin, 2015). Hong Kong investors are more sophisticated and can better monitor and advise firm management on innovative research

and investments. Therefore, we expect that the implementation of the stock market connection program would promote corporate innovation.

To test our prediction, we utilize a difference-in-differences (DiD) and propensity-score matching research design. The numbers of different patents granted by the China National Intellectual Property Administration (CNIPA) are used as our measures of corporate innovation. Connected (non-connected) firms, on which Hong Kong investors are (are not) allowed to trade on the Shanghai Stock Exchange, are used as the treatment (matched control) firms for our DiD regressions. We find that, compared with the non-connected firms, the connected firms become more innovative after the stock market connection, suggesting that it boosts corporate innovation. Our results and inferences continue to hold under an array of robustness checks using placebo tests as well as alternative sample, matching method, model specifications, and measures of corporate innovation.

To lend credence to the foregoing mechanisms through which the Shanghai-Hong Kong stock market connection promotes innovation, we further conduct several mechanism tests. In accord with the first mechanism that the stock market connection boosts firm innovation via facilitating external financing, we find that financial constraints are reduced following the implementation of the stock market connection, and that the stimulating effect of the stock market connection on corporate innovation is more pronounced for firms with relatively greater financial constraints. Our second mechanism postulates that the stock market connection encourages firm innovation via enhancing the informational feedback effect of stock prices. Consistent with this mechanism, we find that, after the stock market connection, stock prices better reflect firm-specific information, and that firms with highly informative stock prices are more innovative. In line with our third mechanism that the stock market connection spurs corporate innovation via enhancing the monitoring and advising on innovative strategies and activities, we find that sophisticated institutional investments increase after the establishment

of the stock market connection, and that the positive effect of the stock market connection on innovation is stronger for firms that feature more Hong Kong institutional investments. Additional cross-sectional results reveal that the effect of the stock market connection is stronger for non-state-owned firms, firms with few political connections, firms with weak intellectual property rights protection, or firms that are headquartered in non-high-tech economic zones.

Our study makes four main contributions to the literature. First, the degree of innovation is significantly higher in developed countries like U.S. than in many developing countries. The latter are in greater need of innovation to promote economy. Nonetheless, while there is a vast literature examining corporate innovation in the developed U.S. market, evidence on corporate innovation in emerging countries, especially China, is sparse. We contribute to this literature by providing evidence on how the Shanghai-Kong Kong stock market connection stimulates innovation of mainland Chinese listed companies. Given a set of unique institutional features of China, we also show how this stimulating impact on innovation varies for state-owned firms, firms with political connections, firms with weak protection of intellectual property rights, and firms that are in the national high-tech economic zones.

Second, a large body of literature examines how corporate innovation is shaped by various firm characteristics and economic environments such as corporate venture capital (Chemmanur, Loutskina, and Tian, 2014), hedge fund activism (Brav, Jiang, Ma, Tian, 2018), credit supply (Amore, Schneider, and Zaldokas, 2013), institutional stock holdings (Aghion, Van Reenen and Zingales, 2013; Luong, Moshirian, Nguyen, Tian, and Zhang, 2017), independent boards (Balsmeier, Fleming, Manso, 2016), private equity ownership (Ferreira, Manso, and Silva, 2014), intellectual property rights protection (Fang, Lerner, and Wu, 2017), managerial capability (e.g., Custodio, Ferreira, and Matos, 2019; Chemmanur, Kong, Drishnan, and Yu, 2019), long-term managerial compensation plans (Manso, 2011), CEO overconfidence

(Galasso and Simcoe, 2011; Hirshleifer, Low, and Teoh, 2012), and investors' greater tolerance for failure (Manso, 2011; Tian and Wang, 2014). Far less research attention has been paid to how economic policy and regulatory reform promote corporate innovation, especially in developing countries such as China. We fill the gap in the literature by showing that stock market connections with more developed stock markets spur innovations of domestic companies.

Third, there is a paucity of empirical evidence on the real economic consequences of opening an emerging stock market to a more developed stock market. Our study adds to this line of research by providing new evidence on the real effect of the connection of Shanghai stock market with Hong Kong stock market. While the former market is widely regarded as considerably less developed than the latter (e.g., Hung, Wong, and Zhang, 2012; Ke, Rui, and Yu, 2012; Ke, Lennox, and Xin, 2015), both are economically autonomous from each other.

Fourth, our study differs from, but complements, that of Moshirian, Tian, Zhang, and Zhang (2020). Using a sample of 20 economies, they find that stock market liberalization, through permitting foreign investments in domestic stock markets, has a positive impact on technological innovation. While this finding is important, we allow for the possibility that, due to institutional and cultural differences, foreign investors, who are generally less acquainted with domestic stock markets and local institutions, could be either more sophisticated or less sophisticated in stock investments than domestic investors. In contrast, investors from Hong Kong are not only more sophisticated than those in mainland China (e.g., Hung, Wong, and Zhang, 2012; Ke, Lennox, and Xin, 2015), but are also more conversant with the institutional setting and stock markets of mainland China, compared with foreign investors. Therefore, the establishment of the Shanghai-Hong Kong stock market connection in 2014 is a unique setting to explore how the openness of a developing stock market to generally more sophisticated investors in a more developed stock market influences firm innovation. Our study is the first

to investigate this issue. Focusing on mainland China for the study not only obviates the challenge of addressing the endogeneity and unobservable heterogeneity that are associated with cross-country differences (e.g., Fang, Lerner, and Wu, 2017), but also, more importantly, adds to the research on the open-door policies pursued by mainland China, the largest developing economy and the second-largest economy in the globe.

The remainder of this paper proceeds as follows. Section 2 describes our data sources and sample. Section 3 presents our research design and main results. Section 4 conducts further analysis of how the stock market connection affects corporate innovation in various circumstances. Section 5 concludes the paper.

2. Data sources, sample selection, and sample matching

2.1. Data sources and sample selection

In 2014, the China Securities Regulatory Commission (CSRC) reached an agreement with the Hong Kong Securities Regulatory Commission (HKSRC) to launch a stock market connection program, under which Hong Kong investors would be permitted to trade on connected stocks on the Shanghai stock market. The connected stocks consist of (i) stocks whose prices compose the Shanghai Stock Exchange 180 index and 380 index and (ii) stocks listed on both the Shanghai Stock Exchange and the Hong Kong Stock Exchange (i.e., A+H stocks). To study how corporate innovation was affected by the stock market connection, we test how the connected firms' innovation would change in a six-year period surrounding the implementation of the stock market connection in 2014, as compared to the changes in innovations of the non-connected firms. Accordingly, our sample selection starts with the entire population of all firms listed on the Shanghai Stock Exchanges over the years 2011-2016. This sample period covers the three-year pre-connection period (i.e., 2011-2013) and the three-year post-connection period (i.e., 2014-2016). In constructing our corporate innovation variables,

we collect the patent grant data from the website of the Chinese State Intellectual Property Office for the period of 2012-2017.¹ Then, we merge the patent data with the financial and stock market data obtained from the China Stock Market and Accounting Research (CSMAR) database and the Wind database for our sample period. This data merging yields an initial sample of 15,144 firm-year observations.

Further, we do the following refinements on the initial sample. First, we remove financial institutions, because their financial characteristics and financial statements are not comparable to those of non-financial firms (e.g., He, 2016). We also eliminate firms identified by CSRC as suffering financial distress and pending delisting during our sample period. As a result, our sample size drops to 14,295 firm-year observations. We further tease out the mainland listed companies that have issued H shares (i.e., shares traded on the Hong Kong Stock Exchange) or B shares (i.e., shares available only for foreign investors to trade) before or during our sample period. This reduces our sample to 14,151 observations. We exclude observations that have negative retained earnings in 2013, the year before the stock market connection program was implemented. 238 firms that were listed during the three years since the implementation of the stock market connection are also removed. We further delete 97 firms that are listed on both the Shanghai Stock Exchange and the Hong Kong Stock Exchange (namely, A+H companies), since these firms have already had access to finance on the Hong Kong Stock Exchange before its connection with Shanghai Stock Exchange. Finally, we require firm-year observations for which the data required for constructing all regressors for our multivariate analysis are not missing. As a result, our sample comprises 11,742 firm-year observations for 2,158 unique firms across the years 2011-2016.

¹ <http://epub.sipo.gov.cn/gjcx.jsp>. To better establish causality for the impact of the stock market connection, we measure all the independent variables at year t , while the corporate innovation variables are measured at year $t+1$. Therefore, the patent data are collected for the period of 2012-2017, one year leading our sample period of 2011-2016.

2.2. Propensity-score matching of connected firms with non-connected firms

There might exist systematic differences in firm characteristics between connected firms (i.e., firms with stocks on which Hong Kong investors are allowed to trade under the stock market connection program) and non-connected firms (i.e., firms whose stocks are not allowed to be traded by Hong Kong investors). This might confound our analysis which involves essentially the comparison of changes in innovations between connected firms and non-connected firms. To remediate this issue, we employ a nearest-neighborhood propensity-score-matching approach to obtain a sample composed of treatment firms (i.e., connected firms) and matched controlled firms (i.e., non-connected firms).

The difference-in-differences research design requires a non-random assignment of observations into the treatment group and the control group, and that covariates affecting the assignment are balanced between treatment firms and control firms in the absence of the treatment event. Therefore, our matching of firms is based on 2013, the year before the event. We match each treatment firm, without replacement, with a control firm using the closest propensity score. The propensity score is estimated from a logit regression, in which the binary variable (*dum_connect*), indicating whether or not a Chinese listed firm is connected with the Hong Kong Stock Exchange, is regressed on a vector of matching covariates. The covariates include return on assets (*roa*), sales growth (*salesgrowth*), market-to-book ratio (*mb*), firm age (*age*), firm size (*size*), financial leverage (*lev*), Qualified Foreign Institutional Investors' stock ownership (*QFII_totshares*), sales performance (*sales*), political connection (*politic_connect*), ownership attribute (*non-state-owned*), capital expenditures (*capex*), and operating cash flow (*opcash*). All the matching covariates are expected to be related to *dum_connect* and are defined in Appendix. We also include firm dummies, year dummies, and industry dummies in our matching regression so as to ensure that, post matching, we will achieve a non-random classification of treatment groups and control groups. After the propensity-score matching, we

obtain our final sample comprising 6,992 firm-year observations for 1,526 unique firms.

To check the effectiveness of our matching, we conduct both the univariate and multivariate checks of covariate balance. First, for the post-match sample, we run two-sample t-tests and compute standardized bias, as per, e.g., Rosenbaum and Rubin (1985), for each of our matching covariates. Panel A of Table 1 reports the results for the univariate covariate balance check. All the mean differences in the covariates are not statistically significant, with p values all above 10%. The standardized bias is less than 10% for all the covariates. These results indicate that our post-match sample achieves a covariate balance and that our matching substantively reduces differences between the connected firms and the non-connected firms.

To further rest assured the covariate balance, we run the foregoing logit regression based on the post-match sample. Column (1) (Column (2)) of Panel B shows the results from running the logit regression based on the pre-match (post-match) sample. While almost all the covariates have statistically significant coefficients for the pre-match sample, the coefficients for all the covariates become not statistically significant after the matching. This again corroborates the effectiveness of our propensity-score matching.

2.3. Univariate statistics

We measure corporate innovation by three types of patents granted by the China National Intellectual Property Administration (CNIPA). The first is invention patents. These are granted for a new product, a new process, or an improvement, which have new unique functions or utilities for consumers. The second type of patents are product-modeling patents, which are granted for a new solution relating to the shape and/or structure of a product. The third type of patents are product-design patents. These are granted for new product design in respect of the shape, color, and/or pattern of a product, which are aesthetically appealing to customers.

To reduce the potential problem of skewness or outliers, we take the natural logarithm

of one plus the number of granted invention patents (*invention*), of granted product-modeling patents (*modeling*), and of granted product-design patents (*design*), respectively, and use them as our measures of corporate innovation. Panel A of Table 2 tabulates the mean values of the three variables across different industries. Firms in the manufacturing industry have the highest mean of *invention*, *modeling*, and *design*. This suggests that the manufacturing industry is the most innovative. The innovativeness helps manufacturing in China to be eminent and competitive worldwide.

To allay the concern on outliers, we winsorize all the continuous variables, which are explanatory for corporate innovation, at the 1% and 99%, respectively. Panel B of Table 2 reports the mean, standard deviation, minimum values, maximum values, and quartiles of all variables used for our regression analysis. Panel C shows the Spearman correlation matrix among the variables used in the multivariate tests. The values of all the correlation coefficients are less than 50%, suggesting that multicollinearity does not exist to pose a threat to our multivariate tests. We also run the variance inflation factors (VIF) test. The results of this test, not tabulated for brevity, show that VIF values for all independent variables are less than 5, further corroborating that multicollinearity is unlikely to affect our regression analysis (O'Brien, 2007).

3. Research design and main results

3.1. Baseline difference-in-differences regressions

To test the effect of the stock market connection on corporate innovation, we start by using the following difference-in-differences OLS regression model:

$$\begin{aligned} invention(modeling, design)_{i,t+1} = & \alpha_0 + \alpha_1 after_i + \alpha_2 dum_connect_t + \\ & \alpha_3 after_i * dum_connect_t + \sum_k \alpha_{i,t,k} controls_{i,t}^k + YearDummies + IndustryDummies + \varepsilon_{i,t} \end{aligned} \quad (1)$$

The dependent variable is *invention*, *modeling*, or *design*, which are the three proxies for

corporate innovation and are defined previously. To enhance the causality relationship between the stock market connection event and corporate innovation, we measure the dependent variable one-year ahead of all the independent variables. The treatment indicator variable, *dum_connect*, is equal to 1 for a connected firm, of which stocks can be traded by Hong Kong investors, and 0 for a non-connected firm. The time indicator variable, *after*, equals 1 if a firm is in a fiscal year during the post-connection period (i.e., 2014-2016), and 0 if the firm is in the pre-connection period (i.e., 2011-2013).² The variable of interest to our test of research question is the interaction term, *after_i*dum_connect_i*. Its coefficient captures the effect of the stock market connection on corporate innovation for the connected firms relative to the non-connected firms. A larger difference-in-differences estimator (α_3) would indicate greater impact of the stock market connection on corporate innovation.

We include a battery of control variables in model (1), based on previous literature on corporate innovation (e.g., Hirshleifer, Low, and Teoh, 2012; Fang, Lerner, and Wu, 2017; Luong, Moshirian, Nguyen, Tian, and Zhang, 2017). These variables include return on assets (*roa*), sales growth (*salesgrowth*), market-to-book ratio (*mb*), firm age (*age*), firm size (*size*), financial leverage (*lev*), Qualified Foreign Institutional Investors' stock ownership (*QFII_totshares*), capital expenditures (*capex*), sales performance (*sales*), political connection (*politic_connect*), ownership attribute (*non-state-owned*), and operating cash flow (*opcash*), which are defined in Appendix. We further include year dummies and industry dummies in the regression, since corporate innovation tends to vary significantly across years and industries (e.g., Moshirian, Tian, Zhang, and Zhang, 2020).³

² The stock market connection program took effect on 17 November 2014. It is possible that the real effect of the stock market connection takes time to materialize. To better account for this possibility, we re-define our post-event period as 2015-2017, and then re-run model (1) based on this alternative post-event period. We get similar results, which support our prediction that the stock market connection promotes corporate innovation.

³ The variance inflation factors (VIF) values for year dummies and *after_i* are below 5, suggesting that multicollinearity is less of a concern with year dummies and *after_i*. Therefore, we include year dummies in the regression to control for the time effect for the post-connection period (i.e., 2014-2016) and pre-

Table 3 reports the OLS regression results for model (1). In the three columns where results for the three innovation measures are tabulated, the coefficients for the interaction terms are all positive and statistically significant at conventional levels. This indicates that, following the stock market connection, innovation of connected firms increases to a larger degree than that of non-connected firms. We interpret this as suggesting that the stock market connection spurs corporate innovation. A one-standard-deviation variation in $after_i * dum_connect_t$ for the *invention (modeling and design)* regression amounts to 0.173 (0.147 and 0.052), which accounts for 69.04% (106.43% and 99.95%) of the full sample mean of *invention (modeling and design)* and hence is economically significant.

3.2. Robustness checks of baseline regression results

3.2.1. Firm-fixed-effects difference-in-differences regression

It is possible that the stock market connection event also caused exogenous changes in some unobserved firm-specific factors that affect corporate innovation. To ease this concern, we run firm-fixed-effects regression for model (1). Because our matching of firms is based on the data of 2013 (the year before the event), there is no time-series variation in our treatment indicator variable, $dum_connect_t$, for our sample. Thus, when including firm-fixed effects in model (1), $dum_connect_t$ is omitted from the regression estimation. To avoid potential confounding effects that an overlapping of regressors may have on our DiD estimators, we first run a firm-fixed effects regression that includes only $dum_connect_i$, $after_t$, $after_i * dum_connect_i$, and year dummies.⁴ The regression results are shown in Columns (1)-(3) of Table 4, with the DiD estimators qualitatively the same as those reported in Table 3. We then run another firm-

connection period (i.e., 2011-2013), respectively. All our main results remain qualitatively identical if we include industry-year interacted dummies, or exclude year dummies, in the regression.

⁴ We do not include industry dummies to avoid their potential multicollinearity with firm-fixed effects. That said, we obtain qualitatively identical results if we include industry dummies in the firm-fixed-effects regression.

fixed-effects regression for model (1) including all the control variables. Results are shown in Columns (4)-(6). They elicit the same inferences as do the results in Table 3. Collectively, our firm-fixed-effects regression results corroborate our conjecture that the stock market connection promotes corporate innovation.

3.2.2. *Tests of parallel trends assumption and time effect*

The parallel trends assumption underlying the difference-in-differences design requires that, in the absence of the treatment event, the outcome variable should exhibit similar trends for both treatment firms and control firms (Roberts and Whited, 2013). To test whether this assumption is tenable for our DiD test, we first compare the year-on-year growth in innovation of the connected firms with that of the non-connected firms for our pre-event sample period (i.e., 2011-2013). The year-on-year growth is measured as: a change in corporate innovation from the previous year to the current year, divided by corporate innovation in the previous year. Results from standard t-tests (un-tabulated) reveal that the growth rates in innovation of the connected firms are not statistically different from those of the non-connected firms for year 2011, year 2012, and year 2013, respectively.

Further, we conduct two multivariate tests of the parallel trends assumption. First, in the spirit of our univariate test of the assumption, we re-run our DiD regression, using 2010 and 2011 (2011 and 2012, and 2012 and 2013) as the pre- and post-treatment periods, respectively. Our un-tabulated results reveal no significant change in innovation for the connected firms relative to the non-connected firms. Second, we augment model (1) by including the interaction terms between year dummies and our treatment indicator variable, $dum_connect_t$, and run this augmented model using OLS regression. Table 5 reports the results. The interaction terms, $after11_i*dum_connect_t$, $after12_i*dum_connect_t$, and $after13_i*dum_connect_t$, all take on a coefficient that is not statistically significant. Collectively, the foregoing results all support the

parallel trends assumption holding for our DiD research design.⁵

The coefficients on the interaction terms, $after14_i*dum_connect_t$, $after15_i*dum_connect_t$, and $after16_i*dum_connect_t$, are all positive and statistically significant, indicating that the stock market connection took effect on corporate innovation in each year of our post-event sample period (i.e., years 2014-2016). The magnitude of the coefficients for $after16_i*dum_connect_t$ is significantly larger than that for $after15_i*dum_connect_t$ (F-stat.=10.48, 9.46, 8.74 and $p=0.003$, 0.003, 0.007 for the *invention*, *modeling*, and *design* regressions, respectively), the latter of which is in turn significantly greater than that for $after14_i*dum_connect_t$ (F-stat.=12.01, 11.73, 6.72 and $p=0.001$, 0.001, 0.01). This suggests that the real effect of the stock market connection on corporate innovation materializes over time.

3.2.3. Placebo test for controlling potential anticipation effect

Before the Shanghai-Hong Kong stock market connection was enforced, some companies might have anticipated this governmental policy and taken advance action to pursue innovative investments and research. This might provide an alternative explanation for our baseline regression results. To alleviate this concern, we re-run the difference-in-differences regression model (1), using 2010-2012 and 2013-2015 as pre- and post-event periods, respectively, so as to test whether there is a foregoing anticipation effect in 2012. In our regression results, tabulated in Panel A of Table 6, there is no statistically significant result for the DiD estimators. This suggests that the anticipation effect is unlikely to explain our baseline results.

⁵ If the stock market connection program were launched for purpose of facilitating financing needs of innovative firms, we should observe an increase in corporate innovation prior to the connection event. This increased innovation might drive both the connection event and the post-event innovation, thereby giving rise to a dynamic type of endogeneity issue. Nonetheless, in our tests of the parallel trends assumption, no significant change in corporate innovation before the connection event rules out the possibility that the dynamic endogeneity drives our baseline results.

3.2.4. *Placebo test using A+H companies as the alternative treatment sample*

Though the connected firms specified by the CSRC encompass firms listed on both the Shanghai Stock Exchange and the Hong Kong Stock Exchange (i.e., A+H companies), their stocks had already been traded by Hong Kong investors before the stock market connection was enforced. Investors in Hong Kong might be conversant with, and have been monitoring, the A+H companies prior to the connection. We use the A+H companies as our alternative treatment sample, and match them, without replacement, with the non-connected firms utilizing the same propensity-score-matching approach as described in Section 2.2. We then re-run our DiD regression based on this new matched sample. Results are reported in Panel B of Table 6. The DiD estimators are not statistically significant, suggesting that the A+H companies do not exhibit an increase in innovation activities that is larger than that of non-connected firms. This, combined with our baseline regression results, substantiate the proposition that, under the connection program, entitling Hong Kong investors to invest in stocks of mainland Chinese firms is the key for stimulating corporate innovation. Also, by using the same control variables as in model (1) for the DiD placebo regression, we further ease the concern that the treatment effect shown in our baseline regression results is driven by potential omitted variable (s).

3.2.5. *Newly added connected firms after the implementation of the stock market connection*

The constituent stocks of the Shanghai Stock Exchange 180 index and 380 index (i.e., SSE 180 stocks and SSE 380 stocks), which compose our treatment sample, change over time. Subject to changes in firm size, stock liquidity, trading volume, growth rate, return on equity, firms may, from year to year, drop in or drop out of the group of SSE 180 stocks and SSE 380 stocks. For our post-event sample period, 238 firms were newly included as SSE 180 stocks or SSE 380 stocks after the implementation of the stock market connection. Using the same propensity-score-matching approach as in Section 2.2, we match each connected firm, with

replacement, with a non-connected firm for year 2013. This results in 720 firm-year observations corresponding with 238 firms. We re-run our DiD regression based on the new sample of observations and report our results in Panel A of Table 7. The DiD estimators all have a positive coefficient that is statistically significant at the 1% level. This supports our inference that the stock market connection enhances corporate innovation.

3.2.6. *Alternative measures of corporate innovation*

We use the number of granted patents to measure corporate innovation in our main tests, as we believe patent grants are powerfully reflective of a firm's degree of innovation. As a robustness check, we account for other dimensions of innovation, including the amount of R&D expenditures and the number of patent applications.⁶ We also use productivity growth (ΔTFP) to measure a firm's innovation outcome. TFP is computed as the natural logarithm of year-on-year change in the firm-level total factor productivity estimated per Olley and Pakes (1996). We run our DiD regression where the dependent variable is replaced with these alternative measures of corporate innovation. The results are reported in Panel B of Table 7. It is shown that the amount of R&D expenditures, the number of patent applications, and productivity growth all increase as a result of the implementation of the stock market connection.

3.2.7. *Alternative matching method*

To reassure no substantive difference between our treatment sample and control sample for the pre-event sample period, we use an automated coarsening k -to- k coarsened exact

⁶ Patents secured by mainland Chinese firms receive far less citations than patents of US firms, not only because the China patents have relatively shorter lifespan, lower impact, and smaller global reach, but also because the google search is banned in mainland China. It takes longer time for the China patents to get cited as compared to the US patents. Since patent citations often lag far behind patent grants for Chinese firms, we do not employ patent citations as the proxy for corporate innovation in our analysis.

matching, as per Iacus, King, and Porro (2012), to re-form our sample for the baseline regression analysis. To this end, we match the treatment firms, without replacement, with the control firms based on the same covariates as we use for the foregoing propensity-score matching. We then use the coarsened-exact-matched sample to run the DiD regression models. The DiD regression results, not tabulated for simplicity, are all quantitatively identical to our baseline results.

3.3. Tests of the channels through which the stock market connection affects corporate innovation

3.3.1. *Does the stock market connection affect firm innovation by facilitating external financing?*

If the openness of the Shanghai stock market to the Hong Kong stock market better satisfies firms' financing needs for innovation, we would expect (i) that financial constraints are relieved after the enforcement of the stock market connection, and (ii) that our baseline regression results would be more pronounced for financially constrained firms, as they have stronger financing needs. We measure financial constraints using the SA index developed by Hadlock and Pierce (2010).

To test the mechanism, we first run a DiD regression of financial constraints on $dum_connect_i$, $after_i$, $after_i * dum_connect_i$, and a range of related control variables (e.g., Winker, 1999). Panel A of Table 8 reports the regression results. We can observe that $after_i * dum_connect_i$ has a negative and statistically significant coefficient, suggesting that financial constraints are eased after the stock market connection. We further test the mechanism by investigating whether the positive influence of the stock market connection on innovation intensifies for firms that have relatively greater financial constraints. To this end, we partition our full sample into two subsamples based on the sample median of the SA index (SA). Then,

we estimate model (1) separately for the two subsamples that are formed based on *SA*. Panel B reports the results. The coefficient on $after_i * dum_connect_i$ is positive and statistically significant for the high-financial-constraint subsample for the *invention*, *modeling*, and *design* regressions, respectively. But the coefficients for $after_i * dum_connect_i$ are not statistically significant for the low-financial-constraint subsample. These findings indicate that the positive impact of the stock market connection is more evident for firms that have financial constraints.⁷ Together, results for the mechanism tests support the notion that the stock market connection affects firm innovation via facilitating external financing.

3.3.2. *Does the stock market connection affect firm innovation by enhancing the informational feedback effect of stock prices?*

The involvement of Hong Kong investors, who are more sophisticated in stock trades, may make stock prices more efficient. The prices may become more informative about the prospects of innovative investments and research, and thereby help innovative research to progress more efficiently. To the extent that the stock market connection enhances the informational feedback effect of stock prices on firm innovation, we predict that (i) stock prices become more informative about firms after the implementation of the stock market connection, and that (ii) the impact of the stock market connection on innovation is stronger for firms with highly informative stock prices. We use stock price synchronicity, as per Morck, Yeung, and Yu (2000), to capture the degree to which stock prices contain value-relevant firm-specific information. Morck, Yeung, and Yu (2000) and Durnev, Morck, Yeung, and Zarowin (2003) show that stock price synchronicity, defined as the R square from asset pricing regressions, is

⁷ We find that bank loan is reduced following the stock market connection, and that the impact of the stock market connection on innovation is more pronounced for firms that borrow a relatively larger amount of loans from banks. These results (not tabulated) also lend support to our stated mechanism that facilitating external financing is the underlying channel through which the stock market connection promotes corporate innovation.

an inverse measure of the amount of firm-specific information reflected in stock prices. The lower the stock price synchronicity, the more value-relevant firm-specific information stock prices contain, and the stronger the informational feedback effect of stock prices is expected to be.

We first regress stock price synchronicity on $dum_connect_i$, $after_i$, $after_i*dum_connect_i$, and a battery of related control variables (e.g., Zhou, Lin, Li, and Cheung, 2019). Panel A of Table 9 reports the results. The coefficient on $after_i*dum_connect_i$ is negative and statistically significant. This result is consistent with our conjecture that, after the stock market connection was implemented, stock price synchronicity becomes lower, and thus stock prices are more informative about firms. To test the moderating effect of stock price synchronicity on the baseline regression results, we split our full sample into two subsamples based on the sample median of stock price synchronicity ($synchr$), and then estimate model (1) separately for the two subsamples. Panel B shows the results. $after_i*dum_connect_i$ has a highly significant, positive coefficient in the low- $synchr$ subsample, but has a statistically nonsignificant coefficient in the high- $synchr$ subsample, for the *invention*, *modeling*, and *design* regressions, respectively. This suggests that the positive influence of the stock market connection on innovation is more evident for firms with highly informative stock prices. This lends support to our view that the stock market connection affects firm innovation via enhancing the informational feedback effect of stock prices.

3.3.3. *Does the stock market connection affect firm innovation via involving more-sophisticated investors' monitoring and advising on firm management?*

Under the stock market connection program, more-sophisticated Hong Kong investors, especially institutional investors, can better monitor and advise firm management on innovative investments and research. If the stock market connection influences firm innovation by

involving more-sophisticated investors' monitoring and advising on firm management, we would expect that sophisticated investors' investments would increase following the implementation of the stock market connection. Also, the positive influence of the stock market connection on corporate innovation should be more pronounced for firms with more Hong Kong institutional investments. We use the Qualified Foreign Institutional Investor (hereafter, QFII) investments to proxy for Hong Kong institutional investments. The QFII system was established in 2002 by China to partially open its stock market to some specified foreign institutional investors.⁸ It was headquartered in Hong Kong. Most QFIIs are also institutional investors in the Hong Kong stock market (e.g., Chen, Ke, and Yang, 2013). Thus, we regard the high-QFII-invested Chinese listed firms as characterized by more Hong Kong institutional investments.

We first do a DiD regression of QFIIs' stock ownership ($QFII_totshares$) on $dum_connect_i$, $after_i$, $after_i*dum_connect_i$, and a series of related control variables (e.g., Timo, Virk, Wang, and Wang, 2019). Panel A of Table 10 reports the regression results. The variable $after_i*dum_connect_i$ takes on a positive, statistically significant coefficient, suggesting that Hong Kong institutional investments increase following the implementation of the stock market connection. We further test the moderating effect of QFIIs' stock ownership on our baseline regression results. To this end, we divide our full sample into two subsamples based on the sample median of stock holdings by QFIIs, and then run model (1) separately for these two subsamples. Panel B reports the results. For all the *invention*, *modeling*, and *design* regressions, the coefficients for $after_i*dum_connect_i$ in the high-QFII-invested subsamples are positive and statistically significant, whereas the coefficients for $after_i*dum_connect_i$ in the low-QFII-invested subsamples are not statistically significant. Thus, the stock market

⁸ QFIIs need to meet specified conditions to enter the mainland Chinese stock markets. For example, institutional investors must have at least five years of experiences in asset management and at least \$5 billion of assets under management for the most recent fiscal year. QFIIs' eligibility for investments in the Chinese securities market is assessed and approved by the China Securities Regulatory Commission (CSRC).

connection has a stronger impact on innovation for the high-QFII-invested firms than for the low-QFII-invested firms. This is consistent with our conjecture that enhancing the monitoring and advising on firm management for innovation is an underlying channel that induces the stock market connection to promote firm innovation.⁹

4. Further analysis

This section conducts further analysis of how the stock market connection affects corporate innovation in various circumstances. In particular, we examine whether the positive effect of the stock market connection on innovation differs for state-owned firms, firms with political connections, firms with weaker protection of intellectual property rights, and firms that are in the national high-tech economic zones. To undertake the tests, we first partition our full sample into two subsamples based on the sample median of the continuous variables for intellectual property rights protection, and on the binary variables for whether a firm is state-owned, whether a firm has political connection, and whether a firm is headquartered in the national high-tech economic zones, respectively. We then run model (1) based on each pair of partitioned subsamples. Table 11 reports the regression results based on the split subsamples, where the results for the regression coefficients on the control variables are omitted for brevity. Below we set forth our cross-sectional analysis in terms of each of the foregoing moderating effects.

4.1. The moderating effect of state-owned property

The promotion and compensation of managers in the Chinese state-owned enterprises,

⁹ To disentangle the second and third mechanisms, documented in Sections 3.3.2 and 3.3.3., from the financing mechanism, we need to investigate the effect of the stock market connection on innovations of Hong Kong listed companies. Unfortunately, we are unable to perform this test because the Hong Kong Intellectual Property Department, which is in charge of approving and granting patents to the Hong Kong listed companies, does not release any patent-related information to the public.

who are often current or former government bureaucrats, are evaluated more by various social and political objectives than by corporate performance (e.g., Fan, Wong, and Zhang, 2007; Chen, Chen, Lobo, and Wang, 2010). As a result, managerial pay in the state-owned enterprises is of low upwards sensitivity to firm performance (e.g., Ke, Rui, and Yu, 2012; Chen, Guan, and Ke, 2013). Thus, managers in state-owned enterprises have low incentives to take short-term risks to undertake innovative projects. Because of relatively high upwards sensitivity of managerial pay to corporate performance in non-state-owned firms, managers therein have strong incentives to pursue innovation for higher sustainable abnormal profits. Consistent with this notion, the significantly negative coefficients on *dum_connect* in our baseline regression results indicate that, prior to the stock market connection, the connected companies (non-connected companies), which are composed more of state-owned (non-state-owned) enterprises, are less (more) innovative. On the other hand, non-state-owned firms generally receive less financial support from local governments and state banks than state-owned firms (e.g., Brandt and Li, 2003; Chen, Chen, Lobo, and Wang, 2010; He, 2016), and are thus more likely to face financial constraints. The stock market connection would help relieve the potential financial constraints for non-state-owned firms, and thereby better incentivize them and enable them to pursue more innovation. Therefore, we expect innovation to be more pronounced for non-state-owned firms than for state-owned firms after the commencement of the stock market connection.

We follow Wang, Wong, and Xia (2008) to define a listed firm as state-owned if its largest ultimate shareholder is a government entity. We divide our sample firms into state-owned firms and non-state-owned firms, respectively, and run model (1) based on these two subsample firms, respectively. Panel B reports the regression results. Consistent with our expectation, the coefficients for *after_i*dum_connect_i* are statistically significant and positive for non-state-owned firms but not for state-owned firms.

4.2. The moderating effect of political connection

Politically connected firms are more likely to receive strong fiscal support from local governments and get preferential treatments in the granting of loans from state banks (e.g., Brandt and Li, 2003; Chen, Chen, Lobo, and Wang, 2010; He, 2016), resulting in a substantially lower likelihood of financial constraints (Cull, Li, Sun, and Xu, 2015), compared with non-politically-connected firms. Furthermore, due to substantial political rents that politically connected firms can earn easier than non-politically-connected firms (Fisman, 2001), the former firms would likely have weaker incentives to make an effort to innovate for their products and services. Therefore, we expect that, after the enforcement of the stock market connection, innovation would increase to a larger extent for non-politically-connected firms than for politically connected firms.

We follow Hung, Wong, and Zhang (2012) to define a listed firm as politically connected if the chairman of board or CEO is a current or former officer of a government entity. We divide our sample firms into politically connected firms and non-politically-connected firms, and run model (1) based on these two types of firms, respectively. Panel C shows the regression results. The results for the *invention*, *modeling*, and *design* regressions are all consistent with our conjecture. Specifically, the stock market connection has a positive, significant effect on non-politically-connected firms but not on politically connected firms.

4.3. The moderating effect of intellectual property rights protection

Firms' incentives to innovate depend critically on the benefits and costs associated with pursuing innovation. The costs and risks of innovating are higher for firms that have lower intellectual property rights protection. By involving more sophisticated investors' monitoring and advising on innovating and associated intellectual property rights protection, the stock market connection reduces the costs of innovating, and thereby increases the profit from

innovation, to a larger extent for firms that have weaker intellectual property rights protection. As such, these firms would be incentivized to innovate more. Accordingly, we predict that the effect of the stock market connection in promoting innovation would be stronger for firms with weaker intellectual property rights protection.

To measure the degree of a firm's intellectual property rights protection, we use the survey-based prefecture-level intellectual property rights indexes, which were published by the Chinese Academy of Social Science. The indexes vary across different provinces and years and are the same for firms in the same province for a given year. We employ these indexes to compute the prefecture intellectual property rights protection scores (*IPR*), as per Fang, Lerner, and Wu (2017), for our sample firms. We then use the sample median of the scores to divide our full sample into the high-*IPR* subsample and the low-*IPR* subsample for our subsample DiD regression analysis. The results in Panel E indicate that the coefficients on the DiD estimators are positive and statistically significant only for the low-*IPR* subsamples. This is consistent with the contention that the stock market connection magnifies its positive effect on innovations for firms that are subject to weak intellectual property rights protection.

4.4. The moderating effect of establishment of national high-tech economic zones

Several national high-tech economic zones were established by China to attract foreign direct investments, foster technological innovation, and thereby promote economic growth. Firms in these high-tech zones enjoy preferential treatments in the corporate income tax rates, land use, talent recruitments, governmental financial subsidies, and simplified administrative procedures for project approval by local governments, among others. These treatments act as an incentive for firms in the high-tech zones to develop technological innovation. Given the plausibly lower financing costs and better monitoring and advising on innovation after the implementation of the stock market connection, firms in the non-high-tech economic zones

should have stronger incentives to pursue innovation, compared with firms in the high-tech zones where they have already been given an incentive to innovate. Accordingly, we expect that the stimulating effect of the stock market connection on innovation is stronger for firms in the non-high-tech economic zones than for those in the high-tech zones.

As with Tian and Xu (2019), we obtain a list of 136 China national high-tech economic zones from the website of the Ministry of Science and Technology, and manually collect the dates on which the high-tech zones were established.¹⁰ Based on this information, we divide our sample firms into those headquartered in the national high-tech economic zones and those headquartered in the non-high-tech zones. Panel F reports our subsample DiD regression results. As expected, the results for the *invention*, *modelling*, and *design* regressions indicate that the coefficients on the DiD estimators are positive and statistically significant for the subsample of firms headquartered in the non-high-tech economic zones, but not for those headquartered in the high-tech economic zones.

5. Conclusion

The Shanghai stock market and the Hong Kong stock market were connected by the Chinese government in 2014. This stock market connection program entitles Hong Kong investors to trade on a group of stocks on the Shanghai stock market. Using a difference-in-differences regression approach, we investigate how this stock market connection affects firm innovation in mainland China. We argue that the stock market connection likely facilitates external financing, enhances the informational feedback effect of stock prices, involves more sophisticated investors' monitoring and advising on firm management, and thereby spurs corporate innovation. Consistent with this argument, we find that corporate innovation increases after the establishment of the stock market connection. This increase is greater for

¹⁰ <http://www.most.gov.cn/gxjscykfq/ldjh/>.

financially constrained firms, firms with highly informative stock prices, and firms that feature more Hong Kong institutional investments. We also find that, following the stock market connection, financial constraints are relieved, stock price informativeness increases, and sophisticated institutional investments increase, thereby explaining why the stock market connection promotes firm innovation. The increased innovation helps firms earn abnormal profits in a long run, which would in turn be beneficial to the investors from Hong Kong who invest in the Shanghai stock market via the stock market connection program.

Further analysis reveals that the positive effect of the stock market connection on innovation is more pronounced for non-state-owned firms, firms with few political connections, firms with weak intellectual property rights protection, or firms that are headquartered in the non-high-tech economic zones. Overall, our study advances understanding of the real effect of the openness of a stock market. Our analysis and results offer insight into how the opening of a developing stock market to a more developed stock market shapes corporate innovation, which should provide helpful implications for countries that are contemplating or planning on opening their stock markets to overseas stock markets.

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Table 1 Propensity-score matching**Panel A: Univariate tests of covariate balance**

Variables	Matching statuses	No. of firm-years	No. of firms	Mean for treatment firms	Mean for control firms	Standard ized bias	t-stat.	p
<i>size</i>	Unmatched	11,742	2158	22.991	21.477	12.170	30.27***	0.000
	Matched	6,992	1526	22.974	22.909	5.300	0.720	0.471
<i>opcash</i>	Unmatched	11,742	2158	0.070	0.043	26.000	4.500***	0.000
	Matched	6,992	1526	0.068	0.073	-0.500	-0.090	0.925
<i>capex</i>	Unmatched	11,742	2158	0.057	0.072	-16.800	-2.730***	0.001
	Matched	6,992	1526	0.058	0.054	5.400	1.250	0.213
<i>QFII_totshares</i>	Unmatched	11,742	2158	0.157	0.121	4.700	4.602***	0.000
	Matched	6,992	1526	0.160	0.181	-2.700	-0.360	0.716
<i>roa</i>	Unmatched	11,742	2158	0.058	0.049	17.000	3.280***	0.000
	Matched	6,992	1526	0.059	0.065	-6.200	-1.243	0.044
<i>lev</i>	Unmatched	11,742	2158	0.422	0.319	55.800	11.46***	0.000
	Matched	6,992	1526	0.421	0.414	3.700	0.530	0.599
<i>age</i>	Unmatched	11,742	2158	1.350	1.345	0.800	2.062**	0.036
	Matched	6,992	1526	1.346	1.368	-3.900	-0.570	0.572
<i>mb</i>	Unmatched	11,742	2158	0.894	0.549	44.300	12.21***	0.000
	Matched	6,992	1526	0.893	0.802	4.260	1.540	0.125
<i>salesgrowth</i>	Unmatched	11,742	2158	0.163	0.242	-20.900	-1.769*	0.046
	Matched	6,992	1526	0.164	0.152	3.100	0.600	0.546
<i>sale</i>	Unmatched	11,742	2158	21.378	19.003	74	2.490	0.014
	Matched	6,992	1526	21.994	21.994	2.400	0.210	0.836
<i>non_state_owned</i>	Unmatched	11,742	2158	0.312	0.274	13.72	2.152	0.032
	Matched	6,992	1526	0.254	0.202	1.960	1.064	0.691
<i>politic_connect</i>	Unmatched	11,742	2158	0.308	0.683	79.200	2.750	0.007
	Matched	6,992	1526	0.296	0.274	2.168	0.010	0.935

Note: This table reports descriptive statistics of the covariates for the sample of connected firms and the sample of non-connected firms. These two types of samples are formed based on propensity-score matching. Specifically, the results of the two-sample tests of mean and of the standardized bias for the covariates are provided for both the pre-match and post-match samples. All the covariates are defined in Appendix. ***, **, * denote the two-tailed statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel B: Multivariate test of covariate balance

Variables	<i>dum_connect</i>	
	Pre-match sample (1)	Post-match sample (2)
<i>roa</i>	5.673*** (3.256)	1.006 (1.241)
<i>salesgrowth</i>	-0.921*** (-3.968)	-0.838 (-1.433)
<i>mb</i>	0.103 (0.778)	0.347 (0.934)
<i>age</i>	-2.549*** (-8.710)	-2.515 (-1.333)
<i>size</i>	2.300*** (3.359)	2.224 (1.008)
<i>lev</i>	-2.415*** (-4.905)	-2.595 (-1.071)
<i>QFII_totshares</i>	-0.125 (-1.520)	-0.152 (-0.806)
<i>capex</i>	-1.859** (-2.069)	-1.182 (-1.020)
<i>opcash</i>	2.898***	3.084

	(3.860)	(0.313)
<i>sales</i>	0.035*	0.014
	(1.928)	(1.132)
<i>politic_connect</i>	0.0377**	-0.026
	(2.022)	(1.359)
<i>non_state_owned</i>	-0.027***	-0.034
	(-3.125)	(-0.832)
constant	-50.60***	-48.99
	(-2.363)	(-1.146)
industry-fixed effects	Yes	Yes
year-fixed effects	Yes	Yes
firm-fixed effects	Yes	Yes
No. of obs.	11,742	6,992
Pseudo R ²	0.37	0.41

Notes: This table reports the logistic regression result for the determinants of the stock market connection. The sample period is 2011-2016. The regression is run for propensity-score matching and involves the full sample of firm-year observations for all mainland Chinese listed firms. The dependent variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. All the determinant variables for whether firms listed in the Shanghai stock market are connected with the Hong Kong stock market are defined in Appendix. Year dummies, industry dummies, and firm dummies are included in regressions but not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. ***, **, * represent the two-tailed statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 2 Univariate statistics**Panel A Statistics of corporate innovation across industries**

Industry types	<i>invention</i>	<i>modeling</i>	<i>design</i>
Agriculture, forestry, animal husbandry, and fishery	1.141	1.113	1.026
Mining	1.136	1.113	1
Manufacturing	5.488	3.441	2.395
Electricity, heat, gas, and water industries	1.053	1.049	1.005
Construction	1.258	1.227	1.038
Wholesale and retail	1.056	1.019	1.013
Transportation	1.098	1.033	1.029
Accommodation and catering	1	1	1
Information technology	1.281	1.135	1.059
Real estate	1.023	1.012	1.001
Leasing and business service	1.053	1.070	1.022
Scientific research and technical service	1.538	1.607	1
Water conservancy, environment, and public facilities management	1.129	1.064	1.015
Resident service	1	1	1
Education	1.032	1.032	1
Health and social work	1.071	1.022	1
Culture, sports, and entertainment	1.192	1.081	1.022
Others	1	1	1

Notes: This table reports the mean values of three corporate innovation variables across industries. These variables are the natural logarithm of one plus the number of granted invention patents (*invention*), the number of granted product-modeling patents (*modeling*), and the number of granted product-design patents (*design*), respectively. The observations are at the firm-year level for the sample period of 2011-2016.

Panel B: Descriptive statistics of variables

Variables	n	Mean	Min.	25%	Median	75%	Max.	Std.
<i>invention</i>	6,992	0.251	0	0	0	0	8.839	0.810
<i>modeling</i>	6,992	0.138	0	0	0	0	9.163	0.625
<i>design</i>	6,992	0.052	0	0	0	0	7.758	0.356
<i>EBIT</i>	6,992	0.056	0.016	0.037	0.052	0.169	0.783	0.748
<i>after</i>	6,992	0.414	0	0	0	1	1	0.493
<i>dum_connect</i>	6,992	0.228	0	0	0	0	1	0.420
<i>roa</i>	6,992	0.049	0.463	0.016	0.039	0.069	0.345	0.795
<i>salesgrowth</i>	6,992	0.232	0.007	0.211	0.534	0.727	1.655	0.609
<i>mb</i>	6,992	1.902	0	0.325	0.566	1.328	14.38	1.473
<i>age</i>	6,992	1.134	0	0.693	1.099	1.792	2.303	0.709
<i>size</i>	6,992	13.87	3.564	13.781	16.810	19.080	23.510	1.610
<i>lev</i>	6,992	0.458	0.051	0.248	0.415	0.592	0.946	0.228
<i>QFII_shares (%)</i>	6,992	0.078	0	0	0	0	34.130	0.548
<i>capex</i>	6,992	0.043	0	0	0.013	0.053	1.217	0.689
<i>opcash</i>	6,992	13.071	1.610	10.030	15.340	18.790	26.420	1.688
<i>SA</i>	6,992	35.570	-0.064	33.630	35.160	36.900	54.140	2.958
<i>indp</i>	6,992	0.169	0	0	0.076	0.247	1.989	0.221
<i>synchr</i>	6,992	-2.560	-4.280	-2.815	-2.276	-1.744	5.956	3.259
<i>top1_ownership</i>	6,992	58.450	1.320	47.140	59.500	60.400	91.800	1.616
<i>high_techzone</i>	6,992	0.019	0	0	0	0	1	0.136
<i>compet1</i>	6,992	1.620	0.065	1.032	1.340	1.519	3.612	1.272
<i>compet2</i>	6,992	7.328	1.470	6.041	7.395	9.716	12.770	1.901
<i>compet3</i>	6,992	10.030	1.628	2.043	8.962	12.453	15.328	2.33
<i>non_state_owned</i>	6,992	0.100	0	0	0	0.333	0.560	0.167
<i>tobin_q</i>	6,992	2.760	0.070	0.973	1.768	2.081	3.460	2.240
<i>state_owned</i>	6,992	0.540	0	0	0	1	1	1.019
<i>IPR</i>	6,992	6.336	0.230	4.950	6.220	7.390	9.264	1.865

<i>politic_connect</i>	6,992	0.296	0	0	0	1	1	0.469
<i>Dual</i>	6,992	0.474	0	0	0	1	1	0.812
<i>sales</i>	6,992	21.782	19.265	20.909	21.677	22.545	26.207	1.231
<i>politic_connect</i>	6,992	0.003	0	0	0	1	1	0.050
<i>TFP</i>	6,992	1.966	0	0	0	10.120	12.510	3.529

Notes: This table tabulates descriptive statistics of the variables used for the difference-in-differences regression tests. The sample period ranges from 2011 to 2016. All the variables are defined in Appendix.

Panel C: Correlation matrix

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>	<i>dum_connect</i>	<i>after</i>	<i>state_owned</i>	<i>IPR</i>	<i>SA</i>
<i>invention</i>	1							
<i>model</i>	0.124***	1						
<i>design</i>	0.378***	0.263***	1					
<i>dum_connect</i>	-0.049***	-0.084***	-0.048***	1				
<i>after</i>	-0.004	-0.031***	-0.032***	-0.022**	1			
<i>state_owned</i>	0.049***	0.083***	0.062***	-0.265***	0.037***	1		
<i>IPR</i>	0.008	0.001	0.024***	-0.138***	-0.038***	0.146***	1	
<i>SA</i>	-0.071***	-0.058***	-0.053***	0.037***	0.021**	-0.130***	-0.003	1

Variables	<i>size</i>	<i>opcash</i>	<i>capex</i>	<i>QFII_totshares</i>	<i>roa</i>	<i>lev</i>	<i>age</i>	<i>mb</i>	<i>salesgrowth</i>
<i>size</i>	1								
<i>opcash</i>	0.000	1							
<i>capex</i>	-0.087***	0.531***	1						
<i>QFII_totshares</i>	0.078***	0.041***	0.025**	1					
<i>roa</i>	-0.054***	0.257***	0.072***	0.080***	1				
<i>lev</i>	0.150***	-0.120***	0.003	0.003	-0.336***	1			
<i>age</i>	0.334***	0.038***	-0.150***	0.004	-0.175***	0.204***	1		
<i>mb</i>	0.391***	-0.045***	0.032***	0.010	-0.207***	0.485***	0.079***	1	
<i>salesgrowth</i>	0.021*	0.100***	0.130***	0.021*	0.169***	0.107***	0.043***	-0.035***	1

Notes: This table presents the results for the Spearman correlations. The correlation matrix involves the variables used for the difference-in-differences regression tests. The sample consists of 6,992 firm-years that cover the years 2011-2016. All the variables are defined in Appendix. ***, **, * represent the two-tailed statistical significance at the 1%, 5%, and 10% levels, respectively, for the correlation coefficients.

Table 3 Baseline difference-in-differences OLS regressions on corporate innovation

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>
	(1)	(2)	(3)
<i>after</i>	-0.507*** (-2.613)	-0.484*** (-3.269)	-0.529*** (-4.051)
<i>after*dum_connect</i>	0.214*** (2.796)	0.235*** (3.405)	0.146** (2.103)
<i>dum_connect</i>	-0.314*** (-4.905)	-0.345*** (-5.568)	-0.284*** (-4.565)
<i>roa</i>	0.741*** (2.967)	0.597*** (2.771)	0.623*** (3.286)
<i>salesgrowth</i>	-0.035 (-1.164)	-0.036 (-1.482)	-0.019 (-0.938)
<i>mb</i>	0.105*** (2.891)	0.037 (0.711)	0.045 (0.704)
<i>age</i>	0.056 (0.436)	0.048 (1.552)	0.067*** (2.826)
<i>size</i>	-0.013 (-0.361)	-0.065 (-0.201)	-0.030* (-1.727)
<i>lev</i>	0.094 (0.070)	0.085 (0.702)	0.133* (1.669)
<i>QFII_totshares</i>	0.018 (1.041)	0.025 (1.495)	0.017 (1.096)
<i>capex</i>	0.096 (0.212)	0.174 (1.265)	0.302* (1.747)
<i>opcash</i>	-0.038 (-0.332)	-0.047 (-0.464)	-0.060 (-1.590)
<i>sales</i>	0.038 (1.037)	0.012 (0.379)	0.016 (0.913)
<i>politic_connect</i>	-0.165 (-1.183)	-0.154 (-1.270)	-0.089 (-1.041)
<i>non_state_owned</i>	0.116*** (4.084)	0.021 (0.889)	0.013 (0.917)
constant	0.582 (0.905)	0.724 (0.553)	0.620** (2.151)
industry-fixed effects	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes
firm-fixed effects	No	No	No
No. of obs.	6,992	6,992	6,992
Adj.R ²	0.127	0.185	0.135
F-stat.	18.510	15.730	12.025

Notes: This table reports the results of difference-in-differences OLS regressions for the impact of the stock market connection on corporate innovation. The sample period covers six years from 2011 to 2016. The dependent variables are three corporate innovation variables: the natural logarithm of one plus the number of granted invention patents (*invention*), the number of granted product-modeling patents (*modeling*), and the number of granted product-design patents (*design*), respectively. The treatment variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, is the variable of interest. It captures the impact of the stock market connection on innovation for the connected firms (*dum_connect* = 1) relative to the non-connected firms (*dum_connect* = 0). All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for brevity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4 Difference-in-differences firm-fixed-effects regressions on corporate innovation

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>	<i>invention</i>	<i>modeling</i>	<i>design</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after*dum_connect</i>	0.060*** (2.8261)	0.082*** (3.390)	0.095*** (2.634)	0.242*** (3.174)	0.307*** (2.914)	0.275* (1.825)
<i>roa</i>				0.737*** (2.802)	0.625*** (3.018)	0.228** (2.126)
<i>salesgrowth</i>				-0.031 (-1.164)	-0.034 (-1.470)	-0.010 (-0.767)
<i>mb</i>				0.113*** (2.891)	0.205 (0.681)	0.194 (0.780)
<i>age</i>				0.024 (0.436)	0.082 (1.520)	0.012 (0.336)
<i>size</i>				-0.013 (-0.361)	-0.012 (-0.039)	0.059 (0.328)
<i>lev</i>				0.073 (0.070)	0.081 (0.659)	0.046 (0.671)
<i>QFII_totshares</i>				0.016 (1.041)	0.023 (1.336)	0.054 (0.813)
<i>capex</i>				0.037 (0.212)	0.053 (1.103)	0.064 (0.832)
<i>opcash</i>				-0.042 (-0.332)	-0.036 (-0.355)	-0.056 (-0.649)
<i>sales</i>				0.019*** (3.218)	0.016*** (2.720)	0.043* (1.925)
<i>politic_connect</i>				-0.029 (-0.168)	-0.017 (-0.810)	-0.012 (-0.295)
<i>non_state_owned</i>				0.066 (1.149)	0.080 (1.236)	0.074 (1.323)
constant	0.325*** (7.070)	0.417*** (6.325)	0.186*** (4.315)	0.524 (1.320)	0.361 (1.540)	0.277 (1.362)
year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
firm-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	6,992	6,992	6,992	6,992	6,992	6,992
Within-R ²	0.170	0.164	0.210	0.190	0.164	0.156
F-stat	27.04	35.14	19.763	25.432	17.831	19.260

Notes: This table reports the results of the difference-in-differences tests, after including firm-fixed effects in the regressions, for the impact of the stock market connection on corporate innovation. The sample period covers six years from 2011 to 2016. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. *dum_connect* equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, is the variable of interest. It captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5 Tests of parallel trends assumption

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>
	(1)	(2)	(3)
<i>after</i>	-0.811*** (-2.795)	-0.611*** (-3.825)	-0.229*** (-3.236)
<i>after11*dum_connect_</i>	0.132 (0.832)	0.237 (1.518)	0.104 (1.156)
<i>after12*dum_connect</i>	0.255 (1.285)	0.338 (0.619)	0.248 (1.272)
<i>after13*dum_connect</i>	0.321 (0.579)	0.375 (3.099)	0.294 (1.421)
<i>after*dum_connect</i>	0.511** (2.059)	0.516** (2.140)	0.381** (2.124)
<i>after14*dum_connect</i>	0.693*** (2.593)	0.627*** (2.439)	0.432* (1.926)
<i>after15*dum_connect</i>	0.771*** (2.706)	0.745** (2.196)	0.569*** (2.794)
<i>after16*dum_connect</i>	0.845** (2.234)	0.812*** (2.482)	0.627** (2.118)
<i>dum_connect</i>	-0.648*** (-6.292)	-0.517*** (-6.040)	-0.202*** (-4.088)
<i>roa</i>	0.262*** (5.543)	0.638*** (3.372)	0.316*** (2.888)
<i>salesgrowth</i>	-0.042 (-1.570)	-0.028 (-1.254)	-0.017 (-1.341)
<i>mb</i>	-0.022 (-0.775)	-0.075 (-0.310)	-0.023 (-1.643)
<i>age</i>	0.060** (2.386)	0.066*** (3.138)	0.016 (0.949)
<i>size</i>	0.064*** (3.571)	-0.038 (-0.257)	0.031*** (3.586)
<i>lev</i>	0.078 (0.928)	0.090 (1.255)	0.014 (0.327)
<i>QFII_totshares</i>	0.035** (2.125)	0.011 (0.846)	0.012 (1.497)
<i>capex</i>	0.196 (1.358)	0.211* (1.763)	0.101 (1.461)
<i>opcash</i>	-0.101 (-0.901)	-0.112 (-1.199)	-0.278 (-0.514)
<i>sales</i>	0.040 (1.071)	0.012 (0.400)	0.017 (0.927)
<i>politic_connect</i>	-0.166 (-1.165)	-0.150 (-1.262)	-0.071 (-1.035)
<i>non_state_owned</i>	0.023*** (4.348)	0.015 (0.654)	0.010 (0.737)
constant	-0.831** (-2.080)	0.368 (1.108)	-0.512*** (-2.670)
F-stat. and its p-value for $\beta(\text{after14*dum_connect}) =$ $\beta(\text{after15*dum_connec})$	12.010 0.003	11.730 0.003	6.720 0.007
F-stat. and its p-value for $\beta(\text{after15*dum_connec})$ $=\beta(\text{after16*dum_connec})$	10.480 0.001	9.460 0.001	8.740 0.010
year-fixed effects	Yes	Yes	Yes
industry-fixed effects	Yes	Yes	Yes

No. of obs.	6,992	6,992	6,992
Adj.R ²	0.12	0.17	0.13
F-stat.	8.600	7.275	10.137

Notes: This table presents the results for the multivariate test of parallel trends assumption. The multivariate test is done based on model (1) augmented by the interaction terms between year dummies and the treatment indicator variable, *dum_connect*. The sample period covers six years from 2011 to 2016. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). *after11*, *after12*, *after13*, *after14*, *after15*, *after16* are the year dummies for years 2011-2016, respectively. For instance, *after11* equals 1 if a firm is in year 2011, and 0 if a firm is in other years over the sample period of 2011-2016. All the other variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6 Placebo tests**Panel A Controlling for potential anticipation effects**

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>
	(1)	(2)	(3)
<i>after</i>	-0.314** (-2.158)	-0.250* (-1.916)	-0.173 (-1.478)
<i>after*dum_connect</i>	0.074 (0.756)	0.053 (0.917)	0.039 (0.977)
<i>dum_connect</i>	0.302*** (4.199)	0.226*** (3.806)	0.153*** (3.515)
<i>roa</i>	1.037*** (4.950)	0.612*** (2.722)	0.404*** (3.020)
<i>salesgrowth</i>	-0.093*** (-2.989)	-0.029 (-1.183)	-0.014 (-0.967)
<i>mb</i>	-0.067** (-2.001)	-0.034 (-0.907)	-0.028* (-1.879)
<i>age</i>	0.039 (1.298)	0.081*** (3.366)	0.098 (0.728)
<i>size</i>	0.015*** (2.617)	-0.017 (-0.104)	0.043*** (3.056)
<i>lev</i>	0.184* (1.873)	0.217* (1.923)	0.154 (1.207)
<i>QFII_totshares</i>	0.031 (1.639)	0.089 (0.607)	0.017 (1.219)
<i>capex</i>	0.211 (1.515)	0.198 (0.903)	0.231 (0.501)
<i>opcash</i>	-0.014 (-0.060)	-0.093 (-0.154)	-0.170 (-0.384)
<i>sales</i>	0.020*** (5.355)	0.065*** (3.205)	0.043** (2.460)
<i>politic_connect</i>	-0.142 (-0.967)	-0.136 (-1.125)	-0.169 (-0.991)
<i>non_state_owned</i>	0.413 (1.470)	0.115*** (5.015)	0.047*** (3.594)
constant	-0.307*** (-6.186)	-0.147 (-0.379)	-0.812*** (-3.631)
year-fixed effects	Yes	Yes	Yes
industry-fixed effects	Yes	Yes	Yes
No. of obs.	6,992	6,992	6,992
Adj.R ²	0.17	0.124	0.136
F-stat.	12.478	10.153	12.640

Notes: This table reports the results from the placebo test, which are done by re-defining the post-event sample period (pre-event sample period) as spanning years 2013-2015 (2010-2012) for estimation of model (1). The sample period covers six years from 2010 to 2015. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-event (pre-event) period of 2013-2015 (2010-2012). The interaction term, *after*dum_connect*, is the variable of interest. It captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B Using the A+H listed firms as the alternative treatment sample

Variables	<i>invention</i>	<i>modeling</i>	<i>design</i>
	(1)	(2)	(3)
<i>after</i>	-0.234 (-0.261)	-0.096 (-0.171)	-0.302 (-0.587)
<i>after*dum_connect1</i>	0.314 (0.006)	0.228 (0.364)	0.154 (0.710)
<i>dum_connect1</i>	-0.234 (-0.404)	-0.136 (-0.046)	-0.258 (-0.525)
<i>roa</i>	0.623* (1.706)	0.307 (1.385)	0.215 (0.845)
<i>salesgrowth</i>	-0.326 (-0.932)	-0.219 (-0.987)	-0.173 (-0.436)
<i>mb</i>	-0.210** (-2.015)	-0.072 (-1.025)	-0.038 (-0.639)
<i>age</i>	0.260 (1.542)	0.234 (1.058)	0.205 (1.042)
<i>size</i>	-0.062 (-0.614)	-0.034 (-0.290)	-0.058 (-0.458)
<i>lev</i>	0.737 (0.778)	0.620 (0.988)	0.515 (1.564)
<i>QFII_totshares</i>	0.105 (1.237)	0.087 (1.614)	0.092 (0.875)
<i>capex</i>	0.203 (0.591)	0.154 (0.651)	0.436 (0.229)
<i>opcash</i>	-0.416 (-1.380)	-0.350 (-1.219)	-0.674 (-1.521)
<i>sales</i>	0.092 (0.145)	0.048 (0.125)	0.032 (1.034)
<i>politic_connect</i>	-0.491 (-1.460)	-0.228 (-1.186)	-0.163 (-1.075)
<i>non_state_owned</i>	0.163*** (3.520)	0.501* (1.873)	0.437 (1.623)
constant	0.361 (0.978)	0.126 (0.542)	0.762 (0.383)
year-fixed effects	Yes	Yes	Yes
industry-fixed effects	Yes	Yes	Yes
No. of obs.	485	485	485
Adj.R ²	0.346	0.130	0.153
F-stat.	11.375	9.824	12.156

Notes: This table reports the results from the placebo test, which are done by using the A+H listed companies as the alternative treatment sample. This sample is re-matched, without replacement, with the control firms of non-connected firms using the propensity-score-matching approach elaborated in Section 2.2. The sample period covers years 2011-2016. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect1*, equals 1 if a firm is an A+H listed firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect1*, is the variable of interest. It captures the impact of the stock market connection on innovation for the A+H listed firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7 Alternative sample/measures for running the baseline regressions**Panel A: Newly added connected firms after the implementation of the stock market connection**

Variables	<i>invention</i> (1)	<i>modeling</i> (2)	<i>design</i> (3)
<i>after</i>	-0.266*** (-4.212)	-0.370*** (-7.518)	-0.247*** (-3.482)
<i>dum_connect</i>	-0.262*** (-4.736)	-0.292*** (-5.318)	-0.302*** (-4.823)
<i>dum_connect*after</i>	0.306*** (3.576)	0.252*** (3.234)	0.214*** (2.708)
<i>roa</i>	0.604** (2.317)	0.481*** (2.674)	0.526*** (3.038)
<i>salesgrowth</i>	-0.032 (-1.402)	-0.025 (-0.949)	-0.033 (-1.465)
<i>mb</i>	-0.062*** (-2.822)	0.067 (1.234)	-0.072*** (-3.092)
<i>age</i>	0.075*** (4.105)	0.045** (2.126)	0.091*** (4.418)
<i>size</i>	-0.019 (-0.132)	-0.082 (-0.560)	-0.026 (-0.166)
<i>lev</i>	0.203** (2.242)	0.154 (0.780)	0.183** (2.446)
<i>QFII_totshares</i>	0.011 (0.835)	0.083 (0.615)	0.015 (1.086)
<i>capex</i>	0.156 (1.104)	0.241* (1.868)	0.173 (1.252)
<i>opcash</i>	-0.065 (-1.627)	-0.074* (-1.701)	-0.032 (-1.327)
<i>sales</i>	0.045 (1.207)	0.014 (0.782)	0.017 (0.945)
<i>politic_connect</i>	-0.171 (-1.210)	-0.156 (-1.308)	-0.124 (-1.062)
<i>non_state_owned</i>	0.127*** (4.035)	0.206 (0.889)	0.138 (1.059)
constant	0.283 (1.295)	0.512* (1.769)	0.325 (1.211)
Year-fixed effects	Yes	Yes	Yes
Industry-fixed effects	Yes	Yes	Yes
No. of obs.	720	720	720
Adj.R ²	0.13	0.17	0.16
F-stat.	17.26	18.13	12.95

Notes: This table reports the results of the difference-in-differences OLS regressions for model (1), using an alternative sample. This sample consists of the connected firms that are newly added after the enforcement of the stock market connection program, and of the control firms that are re-matched, with replacement, with the newly added connected firms using the propensity-score-matching approach elaborated in Section 2.2. The sample period covers years 2011-2016. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, is the variable of interest. It captures the impact of the stock market connection on innovation for the newly-added connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B Alternative measures of corporate innovation

Variables	<i>app_invention</i>	<i>app_modeling</i>	<i>app_design</i>	<i>RDexp</i>	ΔTFP
	(1)	(2)	(3)	(4)	(5)
<i>after</i>	-0.362*** (-4.289)	-0.208*** (-3.143)	-0.155 (-1.438)	-0.539*** (-6.902)	-0.181* (-1.922)
<i>after*dum_connect</i>	0.069* (1.777)	0.061** (2.182)	0.037*** (2.929)	0.137*** (2.679)	0.053** (2.287)
<i>dum_connect</i>	-0.357*** (-4.465)	-0.223*** (-3.572)	-0.120*** (-3.337)	-0.086** (-2.156)	-0.034 (-1.604)
<i>roa</i>	0.135*** (5.099)	0.651*** (3.152)	0.371*** (3.114)	0.757*** (3.099)	0.707*** (2.747)
<i>salesgrowth</i>	-0.088*** (-2.821)	-0.029 (-1.189)	-0.015 (-1.037)	-0.103*** (-3.574)	-0.207*** (3.623)
<i>mb</i>	-0.078** (-2.329)	-0.020 (-0.746)	-0.027* (-1.751)	-0.074 (-0.240)	-0.421*** (-5.973)
<i>age</i>	0.089*** (2.728)	0.072*** (2.837)	0.016 (1.109)	0.068** (2.258)	0.024 (1.034)
<i>size</i>	0.149*** (7.202)	0.064 (1.039)	0.034*** (3.651)	0.066*** (3.467)	0.369*** (8.501)
<i>lev</i>	0.135 (1.348)	0.115 (1.473)	0.021 (0.465)	0.062 (0.671)	0.193 (1.195)
<i>QFII_totshares</i>	0.386** (2.300)	0.239* (1.824)	0.035 (0.469)	0.173*** (3.145)	0.274 (1.278)
<i>capex</i>	0.206 (1.577)	0.201** (1.966)	0.064 (0.108)	0.303*** (1.033)	0.146*** (4.146)
<i>opcash</i>	0.052 (0.121)	0.015 (0.435)	0.052 (0.269)	0.031 (0.772)	0.065 (0.236)
<i>sales</i>	0.195 (1.176)	0.022 (0.170)	0.062 (0.835)	0.128 (0.834)	0.011 (1.203)
<i>politic_connect</i>	-0.123*** (-3.724)	-0.030 (-1.167)	-0.023 (-1.519)	-0.064** (-2.116)	-0.381 (-1.095)
<i>non_state_owned</i>	0.123*** (3.724)	0.030 (1.167)	0.023 (1.519)	0.064** (2.116)	0.093* (1.848)
constant	-0.281*** (-5.982)	0.037 (0.102)	-0.713*** (-3.368)	0.523 (1.204)	-0.126*** (-8.499)
industry-fixed effects	Yes	Yes	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes	Yes	Yes
No. of obs.	6,992	6,992	6,992	6,992	6,992
Adj.R ²	0.172	0.184	0.131	0.248	0.322
F-stat.	16.456	15.676	13.489	26.21	35.39

Notes: This table reports the results of the difference-in-differences OLS regressions for model (1), using alternative measures of corporate innovation. The sample period covers years 2011-2016. The dependent variables, *app_invention*, *app_modeling*, and *app_design*, are the natural logarithm of one plus the number of applied invention patents, the number of applied product-modeling patents, and the number of applied product-design patents, respectively. The other two dependent variables, *RDexp* and ΔTFP , are research and development expenditures and change in firm-level total factor productivity, respectively, and the latter is estimated per Olley and Pakes (1996). The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, is the variable of interest. It captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8 Tests of mechanism: Does the stock market connection affect corporate innovation via facilitating external financing?

Panel A: Are financial constraint relieved after the stock market connection?

Variables	SA
<i>after</i>	-0.165*** (-3.124)
<i>after*dum_connect</i>	-0.385*** (-5.126)
<i>dum_connect</i>	-0.539*** (-2.720)
<i>roa</i>	0.426*** (3.025)
<i>salesgrowth</i>	0.014 (1.147)
<i>size</i>	0.229*** (3.025)
<i>lev</i>	0.556*** (2.540)
<i>capex</i>	-0.230 (-0.859)
<i>opcash</i>	0.625*** (2.601)
<i>sales</i>	-0.308 (-0.436)
<i>politic_connect</i>	-0.086 (-1.473)
<i>non_state_owned</i>	0.182*** (4.167)
constant	-0.448*** (-2.715)
<i>Year-fixed effects</i>	yes
<i>Industry-fixed effects</i>	yes
No. of obs.	6,992
Adj.R ²	0.256
F-stat.	18.6

Notes: This table reports the results of the difference-in-differences OLS regressions for whether financial constraints are reduced after the stock market connection. The sample period covers years 2011-2016. The dependent variable is financial constraints (*SA*), which are measured by the *SA* index developed by Hadlock and Pierce (2010). A higher value of *SA* indicates a larger extent of financial constraints. The treatment variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on financial constraints for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B: The moderating effect of financial constraints

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	High-SA	Low-SA	High-SA	Low-SA	High-SA	Low-SA
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.435*** (-3.125)	-0.624*** (-4.833)	-0.405*** (-2.703)	-0.526*** (-2.539)	-0.254* (-1.836)	-0.263*** (-2.715)
<i>after*dum_connect</i>	0.216*** (3.024)	0.237 (0.590)	0.208*** (3.140)	0.137 (0.444)	0.126** (2.173)	0.135 (1.128)
<i>dum_connect</i>	-0.343*** (-4.206)	-0.160 (-0.851)	-0.253*** (-3.527)	-0.098 (-1.238)	-0.136*** (-2.918)	-0.125 (-1.269)
<i>roa</i>	0.131*** (3.674)	0.234*** (3.420)	0.329* (1.828)	0.861*** (2.974)	0.272 (1.646)	0.243 (1.219)
<i>salesgrowth</i>	-0.040 (-1.237)	-0.026 (-0.699)	-0.098 (-0.657)	-0.039 (-1.242)	-0.094 (-0.927)	-0.016 (-0.784)
<i>mb</i>	-0.213 (-0.685)	0.134 (0.708)	-0.428 (-0.379)	0.112 (1.125)	-0.256 (-1.004)	-0.243 (-1.404)
<i>age</i>	0.051 (1.337)	0.038 (1.435)	0.052 (1.580)	0.043 (1.4123)	0.015 (0.746)	0.046 (0.308)
<i>size</i>	0.012*** (2.306)	0.023 (1.543)	0.025 (1.182)	0.047 (0.695)	0.048*** (2.509)	0.027 (1.248)
<i>lev</i>	-0.033 (-0.954)	0.026* (1.880)	-0.032 (-0.659)	0.031*** (2.780)	-0.020 (-0.615)	0.096 (1.327)
<i>QFII_totshares</i>	0.024 (1.138)	0.051* (1.864)	0.018 (1.490)	0.058 (0.741)	0.017* (1.820)	0.059 (1.415)
<i>capex</i>	0.717** (2.621)	-0.811 (-0.490)	0.572** (2.700)	-0.396 (-0.553)	0.234** (1.972)	-0.215 (-1.014)
<i>opcash</i>	0.048 (0.290)	0.022 (1.164)	0.051 (0.827)	0.084 (1.277)	0.036 (0.937)	0.021 (0.843)
<i>sales</i>	0.047 (1.084)	0.017 (1.245)	0.013 (1.351)	0.088 (1.235)	0.015 (0.972)	0.063 (1.173)
<i>politic_connect</i>	-0.750 (-1.470)	-0.403 (-1.314)	-0.916 (-1.283)	-0.291 (-1.350)	-0.692 (-1.361)	-0.407 (-1.325)
<i>non_state_owned</i>	0.140*** (3.206)	0.111** (2.623)	0.190 (1.358)	0.734 (1.164)	0.107 (0.721)	0.660 (1.322)
constant	-0.241*** (-3.925)	-0.173 (-1.362)	-0.325 (-0.693)	-0.614 (-1.352)	-0.101*** (-3.321)	-0.471 (-1.526)
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	3,840	3,152	3,840	3,152	3,840	3,152
Adj.R ²	0.151	0.097	0.176	0.150	0.134	0.125
F-stat.	9.305	8.605	9.763	10.143	8.626	9.543

Notes: This table reports the results from testing the moderating effect of financial constraints on the regression estimations of model (1). The sample period spans years 2011-2016. The moderator variable is financial constraints (*SA*), which are measured by the *SA* index developed by Hadlock and Pierce (2010). A higher value of *SA* indicates a larger extent of financial constraints. The difference-in-differences regressions are run separately in the low-*SA* subsample and the high-*SA* subsample, which are split based on the full-sample median of *SA*. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 9: Test of mechanism: Does the stock market connection impact firm innovation via enhancing the informational feedback effect of stock prices?

Panel A: Does the stock price synchronicity become lower after the stock market connection?

Variables	<i>synchr</i>
<i>after</i>	-0.521*** (-3.424)
<i>after*dum_connect</i>	-0.217*** (-2.501)
<i>dum_connect</i>	0.037 (0.903)
<i>roa</i>	-0.122*** (-3.447)
<i>salesgrowth</i>	0.350*** (3.146)
<i>size</i>	-0.539*** (-2.514)
<i>QFII_totshares</i>	-0.116*** (-4.003)
<i>lev</i>	0.267*** (2.418)
<i>capex</i>	-0.054 (-0.872)
<i>opcash</i>	-0.315** (-2.117)
<i>sales</i>	-0.047 (-1.352)
<i>politic_connect</i>	0.475 (1.203)
<i>non_state_owned</i>	-0.340*** (-2.829)
constant	0.183*** (3.280)
Year-fixed effects	Yes
Industry-fixed effects	Yes
Adj.R ²	0.18
No. of obs.	6,992
F-stat.	12.69

Notes: This table reports the results of the difference-in-differences OLS regressions for whether stock price synchronicity is reduced after the stock market connection. The sample period covers years 2011-2016. The dependent variable is stock price synchronicity (*synchr*), which is measured per Morck, Yeung and Yu (2000). A higher value of *synchr* indicates a larger extent of stock price synchronicity and a lower level of stock price informativeness. The treatment variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after* dum_connect*, captures the impact of the stock market connection on stock price synchronicity for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B: The moderating effect of stock price synchronicity

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	Low- <i>synchr</i>	High- <i>synchr</i>	Low- <i>synchr</i>	High- <i>synchr</i>	Low- <i>synchr</i>	High- <i>synchr</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.620*** (-8.206)	-0.253 (-1.327)	-0.436*** (-4.053)	-0.252 (-1.328)	-0.180*** (-2.870)	-0.274 (-1.235)
<i>after*dum_connect</i>	0.374*** (3.228)	0.084 (0.729)	0.255*** (2.766)	0.130 (1.125)	0.128** (2.109)	0.026 (0.915)
<i>dum_connect</i>	-0.271*** (-3.106)	-0.362*** (-4.137)	-0.251*** (-2.848)	-0.329*** (-3.541)	-0.271*** (-3.526)	-0.186*** (-2.904)
<i>roa</i>	1.365*** (4.186)	1.724*** (3.215)	0.670*** (3.148)	0.526 (1.504)	0.419*** (2.827)	0.104 (0.826)
<i>salesgrowth</i>	-0.036 (-1.296)	-0.024 (-0.130)	-0.018 (-0.767)	-0.067 (-0.950)	-0.068 (-0.558)	-0.053 (-1.313)
<i>mb</i>	-0.065* (-1.689)	0.052 (0.977)	-0.015 (-0.447)	0.012 (0.298)	-0.024 (-1.285)	-0.039 (-1.423)
<i>age</i>	0.045 (1.604)	0.049 (0.762)	0.063*** (2.655)	0.018 (0.346)	0.011 (0.849)	-0.066 (-0.212)
<i>size</i>	0.037* (1.780)	0.013*** (3.603)	0.046 (0.262)	0.081 (0.027)	0.022** (2.256)	0.060*** (3.328)
<i>lev</i>	0.026 (1.206)	0.036 (0.186)	0.094 (1.152)	0.012 (0.743)	0.038 (0.083)	0.074 (0.789)
<i>QFII_totshares</i>	0.045** (2.372)	0.014 (0.415)	0.012 (0.879)	0.013 (0.487)	0.059 (0.646)	0.027* (1.674)
<i>capex</i>	0.186 (0.110)	0.964** (2.492)	0.183 (1.140)	0.329 (1.210)	0.647 (0.090)	0.428** (2.389)
<i>opcash</i>	-0.025 (-0.209)	-0.024 (-0.703)	-0.063 (-0.617)	-0.015 (-0.539)	-0.019 (0.324)	-0.014 (-0.083)
<i>sales</i>	0.032 (0.897)	0.039 (0.778)	0.014 (0.468)	0.015 (0.044)	0.016 (0.966)	0.033 (0.132)
<i>politic_connect</i>	-0.196 (-1.097)	-0.110 (-0.446)	-0.142 (-0.933)	-0.150 (-0.747)	-0.086 (-1.004)	-0.065 (-0.545)
<i>non_state_owned</i>	0.097*** (2.983)	0.157*** (2.667)	0.097 (0.350)	0.028 (0.583)	0.073 (0.463)	0.057* (1.982)
constant	-0.411 (-0.896)	-0.258*** (-2.901)	0.278 (0.715)	0.142 (0.197)	-0.359 (-1.643)	-0.127*** (-2.952)
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	5,005	1,987	5,005	1,987	5,005	1,987
Adj.R ²	0.10	0.17	0.13	0.12	0.15	0.134
F-stat.	7.79	9.85	10.17	8.962	9.815	12.820

Notes: This table reports the results from testing the moderating effect of stock price synchronicity on the regression estimations of model (1). The sample period spans years 2011-2016. The moderator variable is stock price synchronicity (*synchr*), which is measured per Morck, Yeung, and Yu (2000). A higher value of *synchr* indicates a larger extent of stock price synchronicity and a lower level of stock price informativeness. The difference-in-differences regressions are run separately in the low-*synchr* subsample and the high-*synchr* subsample, which are split based on the full-sample median of *synchr*. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 10: Test of mechanism: Does the stock market connection affect firm innovation via involving more-sophisticated investors' monitoring and advising on firm management?

Panel A: Do QFIIs' stock holdings and other institutional shareholding increase after the stock market connection?

Variables	<i>QFII_totshares</i> (1)
<i>after</i>	0.015** (2.083)
<i>after*dum_connect</i>	0.016* (1.953)
<i>dum_connect</i>	-0.024** (-2.209)
<i>roa</i>	0.013*** (3.547)
<i>salesgrowth</i>	-0.059** (-2.125)
<i>mb</i>	-0.032*** (-2.469)
<i>age</i>	0.457 (1.285)
<i>size</i>	0.018*** (2.640)
<i>lev</i>	-0.096* (-1.818)
<i>capex</i>	-0.087 (-1.359)
<i>opcash</i>	0.024*** (2.528)
<i>sale</i>	0.029 (0.451)
<i>politic_connect</i>	-0.042*** (-2.628)
<i>non_state_owned</i>	0.015 (0.930)
constant	-0.373*** (-2.604)
Industry-fixed effects	Yes
year-fixed effects	Yes
No. of obs.	6,992
Adj.R ²	0.15
F-stat.	9.78

Notes: This table reports the results of the difference-in-differences OLS regressions for whether QFIIs' stock holdings and other institutional shareholdings increase after the stock market connection. The sample period covers years 2011-2016. The dependent variable, *QFII_totshares*, is the shares held by foreign qualified institutional investors (QFIIs), divided by total shares outstanding, for a firm at the end of a year. The treatment variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on QFIIs' stock ownership for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B: The moderating effect of QFIIs' stock ownership

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	High-QFII	Low-QFII	High-QFII	Low-QFII	High-QFII	Low-QFII
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.543*** (-3.126)	-0.109*** (-2.734)	-0.420*** (-4.156)	-0.829** (-2.143)	-0.177*** (-3.291)	-0.492* (-1.785)
<i>after*dum_connect</i>	0.271*** (3.296)	0.420 (0.754)	0.266*** (3.209)	0.304 (1.435)	0.183** (2.126)	0.280 (1.258)
<i>dum_connect</i>	-0.325*** (-2.848)	-0.318 (-0.832)	-0.254*** (-4.156)	-0.295 (-1.370)	-0.176*** (-3.515)	-0.373 (-1.428)
<i>roa</i>	1.261*** (5.460)	1.750 (0.867)	0.656*** (3.290)	0.148 (0.743)	0.253*** (2.584)	0.234** (2.026)
<i>salesgrowth</i>	-0.031 (-1.134)	-0.078 (-0.455)	-0.018 (-0.795)	-0.016 (-0.751)	-0.012 (-0.887)	-0.014 (-1.584)
<i>mb</i>	-0.019 (-0.637)	0.013 (0.569)	-0.025 (-0.103)	0.098 (0.046)	-0.023 (-1.615)	0.015 (0.119)
<i>age</i>	0.035 (1.364)	0.036** (2.170)	0.048** (2.169)	0.029* (1.948)	0.032 (0.256)	0.082 (0.971)
<i>size</i>	0.065*** (3.582)	0.024 (1.217)	0.065 (0.426)	0.031 (1.351)	0.030*** (3.293)	0.078 (0.929)
<i>lev</i>	0.060 (0.913)	0.236 (0.697)	0.071 (1.148)	0.264 (0.538)	0.065 (0.261)	0.171 (0.573)
<i>capex</i>	0.132 (1.235)	0.345*** (3.168)	0.150 (1.248)	0.274*** (2.762)	0.118* (1.690)	-0.816 (-1.384)
<i>opcash</i>	-0.076 (-0.584)	-0.093 (-0.726)	-0.087 (-1.055)	-0.037 (-0.849)	-0.040 (-0.951)	-0.028 (-1.137)
<i>sales</i>	0.036 (-0.946)	0.095 (0.724)	0.099 (0.318)	0.040 (0.268)	0.015 (0.896)	0.020 (1.053)
<i>politic_connect</i>	-0.183 (-0.872)	-0.632 (-1.264)	-0.179 (-0.952)	-0.761 (-1.130)	-0.261 (-0.875)	-0.239 (-1.105)
<i>non_state_owned</i>	0.135*** (3.124)	0.242 (1.426)	0.428 (1.137)	0.137 (0.854)	0.115 (0.837)	0.243 (0.461)
constant	-0.108*** (-2.624)	0.176 (0.700)	0.325 (0.970)	-0.114 (-0.503)	-0.546*** (-2.759)	-0.203 (-0.154)
Industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	5,749	1,243	5,749	1,243	5,749	1,243
Adj.R ²	0.19	0.16	0.17	0.15	0.13	0.098
F-stat.	8.67	9.78	12.06	12.34	8.843	10.265

Notes: This table reports the results from testing the moderating effect of QFIIs' stock holdings on the regression estimations of model (1). The sample period spans years 2011-2016. The moderator variable is QFIIs' stock holdings (*QFII_totshares*), calculated as the shares held by foreign qualified institutional investors, divided by total shares outstanding, for a firm at the end of a year. High-QFII-invested Chinese listed firms are regarded as characterized by more Hong Kong institutional investments. The difference-in-differences regressions are run separately in the low-*QFII* subsample and the high-*QFII* subsample, which are split based on the full-sample median of *QFII*. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. All the variables are defined in Appendix. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 11: Further cross-sectional analyses of the impact of the stock market connection on corporate innovation

Panel A: Does the stock market connection have stronger impact on non-state-owned firms than on state-owned firms?

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	non-state-owned	state-owned	non-state-owned	state-owned	non-state-owned	state-owned
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.580*** (-3.261)	-0.738*** (-2.640)	-0.624*** (-4.182)	-0.511*** (-3.202)	-0.236*** (-4.175)	-0.473* (-1.824)
<i>after*dum_connect</i>	0.375* (1.977)	0.278 (1.235)	0.269** (2.258)	0.150 (1.491)	0.191* (1.876)	0.174 (0.853)
<i>dum_connect</i>	-0.507*** (-4.310)	-0.425*** (-3.521)	-0.377*** (-4.143)	-0.610 (-0.736)	-0.972*** (-3.618)	-0.653 (-1.275)
controls	Yes	Yes	Yes	Yes	Yes	Yes
constant	-0.149** (-1.853)	-0.185** (-2.107)	-0.426 (-0.840)	-0.934 (-1.283)	-0.826*** (-2.877)	-0.815* (-1.904)
industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	4,801	1,199	4,801	1,199	4,801	1,199
Adj.R ²	0.13	0.12	0.18	0.162	0.140	0.175
F-stat.	8.62	9.25	11.38	9.73	10.55	9.71

Notes: This table reports the results from testing whether the regression estimations of model (1) vary across state-owned firms versus non-state-owned firms. The sample period spans years 2011-2016. The moderator variable is a binary variable indicating whether a mainland Chinese listed firm is state-owned (i.e., whether its largest ultimate shareholder is a government entity). The difference-in-differences regressions are run separately for the state-owned subsample and the non-state-owned subsample. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. The results for the regression coefficients on the control variables are omitted for brevity. Industry dummies and year dummies are included in all the regression but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B: Does the stock market connection have stronger impact on firms with few political connection?

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	Politically connected	Non-politically connected	Politically connected	Non-politically connected	Politically connected	Non-politically connected
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.126*** (-2.553)	-0.874*** (-3.019)	-0.761*** (-2.864)	-0.438*** (-3.170)	-0.657*** (-4.291)	-0.194** (-2.601)
<i>after*dum_connect</i>	0.236 (1.2761)	0.156** (2.213)	0.137 (0.992)	0.185** (2.143)	0.346 (0.925)	0.582* (1.943)
<i>dum_connect</i>	-0.380* (-1.793)	-0.344*** (-3.005)	-0.256* (-1.894)	-0.270*** (-3.241)	-0.135 (-1.364)	-0.246*** (-2.513)
controls	Yes	Yes	Yes	Yes	Yes	Yes
constant	-0.194* (-1.856)	-0.168** (-2.239)	-0.724 (-0.985)	0.531 (1.247)	-0.135*** (-2.691)	-0.428 (-1.464)
industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,195	2,410	1,195	2,410	1,195	2,410

Adj.R ²	0.170	0.15	0.16	0.11	0.14	0.13
F-stat.	11.36	8.78	10.27	8.94	8.35	9.65

Notes: This table reports the results from testing whether the regression estimations of model (1) vary across politically connected firms versus non-politically-connected firms. The sample period spans years 2011-2016. The moderator variable is a binary variable indicating whether a mainland Chinese listed firm is politically connected (i.e., whether the chairman of board or CEO in the listed firm is a current or former officer of a government entity). The difference-in-differences regressions are run separately for the politically-connected subsample and the non-politically-connected subsample. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. The results for the regression coefficients on the control variables are omitted for brevity. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel C: Does the stock market connection have stronger impact on firms with weaker intellectual property rights protection?

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	High-IPR	Low-IPR	High-IPR	Low-IPR	High-IPR	Low-IPR
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.153 (-1.094)	-0.289* (-1.821)	-0.214** (-2.169)	-0.237** (-2.533)	-0.162 (-0.927)	-0.128* (-1.935)
<i>after*dum_connect</i>	0.127 (0.854)	0.326*** (3.258)	0.258 (1.263)	0.236** (2.125)	0.161 (0.843)	0.127* (1.860)
<i>dum_connect</i>	-0.477*** (-3.123)	-0.504*** (-2.620)	-0.316*** (-2.347)	-0.343*** (-3.248)	-0.202*** (-3.160)	-0.143** (-2.135)
controls	Yes	Yes	Yes	Yes	Yes	Yes
constant	-1.365 (-1.147)	-1.729** (-2.364)	0.465 (0.826)	-0.826 (-0.943)	-0.727* (-1.928)	-1.134*** (-2.837)
industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	1,859	2,713	1,859	2,713	1,859	2,713
Adj.R ²	0.135	0.126	0.150	0.174	0.150	0.124
F-stat.	11.33	9.62	10.73	8.95	10.62	12.81

Notes: This table reports the results from testing the moderating effect of intellectual property rights protection on the regression estimations of model (1). The sample period spans years 2011-2016. The moderator variable is intellectual property rights protection (*IPR*), which is measured per Fang, Lerner, and Wu (2017). The difference-in-differences regressions are run separately in the low-*IPR* subsample and the high-*IPR* subsample, which are split based on the full-sample median of *IPR*. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. The results for the regression coefficients on the control variables are omitted for brevity. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel D: Does the stock market connection have stronger impact on firms that in non-high-tech economic zones?

Variables	<i>invention</i>		<i>modeling</i>		<i>design</i>	
	High-tech zones	Non-high-tech zones	High-tech zones	Non-high-tech zones	High-tech zones	Non-high-tech zones
	(1)	(2)	(3)	(4)	(5)	(6)
<i>after</i>	-0.153*	-0.553***	-0.634	-0.405***	0.247	-0.138***
	(-1.846)	(-8.124)	(-1.299)	(-7.819)	(0.650)	(-4.723)
<i>after*dum_connect</i>	0.247	0.326***	0.570	0.261***	0.245	0.125**
	(0.537)	(3.719)	(0.836)	(3.917)	(0.691)	(2.514)
<i>dum_connect</i>	0.253	-0.346***	-0.602	-0.435***	-0.751*	-0.142***
	(0.164)	(-5.291)	(-0.743)	(-5.301)	(-1.896)	(-4.035)
controls	Yes	Yes	Yes	Yes	Yes	Yes
constant	-0.126	-0.853**	0.682	0.225	0.391	-0.603***
	(-0.354)	(-2.168)	(1.516)	(0.643)	(1.405)	(-3.24)
industry-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	124	6,868	124	6,868	124	6,868
Adj.R ²	0.22	0.13	0.19	0.177	0.156	0.14
F-stat.	8.56	10.36	9.36	8.94	10.56	12.36

Notes: This table reports the results from testing whether the regression estimations of model (1) vary across firms in the national high-tech economics zones versus firms in the non-high-tech economics zones. The sample period spans years 2011-2016. The moderator variable is a binary variable indicating whether a mainland Chinese listed firm is headquartered in the national high-tech economics zones. The difference-in-differences regressions are run separately in the high-tech-zones subsample and the low-high-tech-zones subsample. The dependent variables are the corporate innovation variables: *invention*, *modeling*, and *design*, respectively. The treatment indicator variable, *dum_connect*, equals 1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise. *after* equals 1 (0) if a firm is in the post-stock-connection (pre-stock-connection) period (i.e., 2014-2016 (2011-2013)). The interaction term, *after*dum_connect*, captures the impact of the stock market connection on innovation for the connected firms relative to the non-connected firms. The results for the regression coefficients on the control variables are omitted for brevity. Industry dummies and year dummies are included in all the regressions but are not reported for simplicity. *t*-statistics in parentheses are based on robust standard errors clustered by firm. *, **, and *** indicate the two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Summary of variable definitions

Variables	Definitions
<i>invention</i>	the natural logarithm of one plus the number of invention patents granted to a listed firm by the China National Intellectual Property Administration in a year.
<i>modeling</i>	the natural logarithm of one plus the number of product-modeling patents granted to a listed firm by the China National Intellectual Property Administration in a year.
<i>design</i>	the natural logarithm of one plus the number of product-design patents granted to a listed firm by the China National Intellectual Property Administration in a year.
<i>dum_connect</i>	1 if a firm is a connected firm on which Hong Kong investors are entitled to trade under the stock market connection program, and 0 otherwise.
<i>after</i>	1 if a firm is in the post-stock-connection period (i.e., 2014-2016), and 0 if a firm is in the pre-stock-connection period (i.e., 2011-2013).
<i>size</i>	the natural logarithm of total assets for a firm at the end of a year.
<i>lev</i>	total liabilities, divided by total assets, for a firm at the end of a year.
<i>roa</i>	net income in a year, divided by total assets at the end of a year, for a firm.
<i>opcash</i>	net operating cash flow in a year, divided by total assets at the end of a year, for a firm.
<i>mb</i>	the market value of equity, divided by the book value of equity, for a firm at the end of a year.
<i>QFII_totshares</i>	the shares held by foreign qualified institutional investors (QFIIs), divided by total shares outstanding, for a firm at the end of a year.
<i>SA</i>	a financial constraint index (SA) developed by Hadlock and Pierce (2010). $SA = -0.737 * size + 0.043 * size^2 - 0.040 * age$, where <i>size</i> is the natural logarithm of total assets at the end of a year, and <i>age</i> is the number of years since a firm was listed on the Shanghai Stock Exchange.
<i>capex</i>	capital expenditures in a year, scaled by total assets at the end of a year, for a firm.
<i>salesgrowth</i>	the difference between sales revenue for the current year and that for the previous year, divided by sales revenue for the previous year, for a firm.
<i>tobin_q</i>	the ratio of the market value of assets to the book value of assets for a firm at the end of a year.
<i>EBIT</i>	earnings before taxes and interests in a year, divided by total assets at the end of a year, for a firm.
<i>age</i>	the natural logarithm of the number of years since a firm was listed on the Shanghai stock market.
<i>non_state_owned</i>	1 if a mainland Chinese listed firm is non-state-owned (i.e., if its largest ultimate shareholder is a non-government entity), and 0 otherwise.
<i>indp</i>	the number of independent non-executive directors, divided by the total number of directors, for the board of a firm at the end of a year.
<i>top1_ownership</i>	the shares held by the largest shareholder, divided by the total shares held by all investors, for a firm in a year.
<i>dual</i>	1 if the CEO holds the position of the chair of the board for a firm in a year, and 0 otherwise.
<i>synchr</i>	stock price synchronicity, which is measured per Morck, Yeung and Yu (2000). It is equal to $\log((R^2/(1-R^2)))$, where R^2 is derived from the market model of regressing daily stock returns of a firm on the daily stock market index of the country for a year.
<i>politic_connect</i>	1 if a listed firm is politically connected (i.e., if the chairman of board or CEO in a listed firm is a current or former officer of a government entity), and 0 otherwise.
<i>IPR</i>	intellectual property rights protection, which is measured per Fang, Lerner, and Wu (2017).
<i>high_techzone</i>	1 if a listed firm is headquartered in the national high-tech economics zones, and 0 otherwise.
<i>TFP</i>	firm-level total factor productivity estimated per Olley and Pakes (1996).
<i>sales</i>	the natural logarithm of a firm's total revenues in a year.